

Self-pinned distances

through unforced scale selection

30 September 2026

This manuscript refines the Hausdorff–packing sufficient condition by separating standard spatial wave packets from the auxiliary angular partitions in the Fourier estimate. The angular marks remain constant on each standard packet label; only the spectral calculation is subdivided more finely. This removes the prescribed midpoint from the scale-selection problem.

The latest refinement first normalizes the profile endpoint and then adapts the preceding interval to the actual tail minimum. A compact interpolation of these choices yields a further improvement for Hausdorff dimensions between $11/10$ and $7/6$.

Scope. The manuscript gives the complete mathematical argument using the explicitly cited established analytic inputs, including both analytic branches and the Borel-set reduction. Internal checks are not external refereeing or Lean kernel verification.

Optimality. The weakest possible condition on the two dimensions is not established. An explicit profile proves sharpness of the upper-slope unforced chain estimate in this method; it is not a planar distance-set counterexample. No unrestricted improvement below the five-quarters threshold is claimed.

The existing Lean challenge and comparator are unchanged. This mathematical manuscript does not complete the unconditional formalization.

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1 Statement and scope

For a Borel set $E \subset \mathbb{R}^2$, write $d = \dim_{\text{H}} E$, $D = \dim_{\text{P}} E$, and $\Delta_y(E) = \{|x - y| : x \in E\}$. All distances are Euclidean and $|\cdot|$ denotes one-dimensional Lebesgue measure. The strongest sufficient curve proved in this manuscript is

$$B_{\text{A}}(d) = \begin{cases} 2d - 1, & 1 < d \leq \frac{11}{10}, \\ 1 + \beta(d - 1), & \frac{11}{10} < d < \frac{7}{6}, \\ \frac{1}{3 - 2d}, & \frac{7}{6} \leq d \leq \frac{5}{4}. \end{cases} \quad (1)$$

Here $\beta(a)$ is the unique solution in $[2a, 1/2]$ of $C_{\text{A}}(a, b) = a$. The function C_{A} is the minimum, over one parameter in $[0, 1]$, of the explicit rational expression (54). Lemma 5.2 proves uniqueness, continuity and the endpoint values $\beta(1/10) = 1/5$, $\beta(1/6) = 1/2$. No numerical optimization hypothesis is used.

Theorem 1.1 (Adaptive sufficient packing condition). *Let $E \subset \mathbb{R}^2$ be Borel. If $1 < d = \dim_{\text{H}} E \leq 5/4$ and $\dim_{\text{P}} E < B_{\text{A}}(d)$, then $|\Delta_y(E)| > 0$ for some $y \in E$.*

The preceding actual-minimum refinement is retained for comparison:

$$B_{\text{M}}(d) = \begin{cases} 2d - 1, & 1 < d \leq \frac{9}{8}, \\ 1 + b_*(d - 1), & \frac{9}{8} < d < \frac{7}{6}, \\ \frac{1}{3 - 2d}, & \frac{7}{6} \leq d \leq \frac{5}{4}. \end{cases} \quad (2)$$

Here $b_*(a)$ is the unique solution in $[2a, 1/2]$ of $V(a, b) = a$, for the explicit rational minimum–maximum expression (38) below. The proof establishes uniqueness, continuity, and the endpoint values $b_*(1/8) = 1/4$, $b_*(1/6) = 1/2$; thus this definition requires no numerical or optimization hypothesis.

Theorem 1.2 (Refined sufficient packing condition). *Let $E \subset \mathbb{R}^2$ be Borel. If $1 < d = \dim_{\text{H}} E \leq 5/4$ and $\dim_{\text{P}} E < B_{\text{M}}(d)$, then $|\Delta_y(E)| > 0$ for some $y \in E$.*

The preceding, simpler sufficient curve remains useful for comparison:

$$B_{\text{U}}(d) = \begin{cases} 2d - 1, & 1 < d \leq \frac{8}{7}, \\ \frac{2 + 2d - 3d^2}{6 - 5d}, & \frac{8}{7} < d \leq \frac{7}{6}, \\ \frac{1}{3 - 2d}, & \frac{7}{6} < d \leq \frac{5}{4}. \end{cases} \quad (3)$$

Theorem 1.3 (Simpler sufficient packing condition). *Let $E \subset \mathbb{R}^2$ be Borel. If $1 < d = \dim_{\text{H}} E \leq 5/4$ and*

$$\dim_{\text{P}} E < B_{\text{U}}(d),$$

then $|\Delta_y(E)| > 0$ for some $y \in E$.

The proof constructs separated compact source and pin subsets of E . A source probability has absolutely continuous pinned distance measures for almost every pin under the selected pin probability. Distance images of Borel sets are analytic, hence Lebesgue measurable.

For example,

$$d = \frac{31}{25} = 1.24 \implies D < \frac{25}{13} = 1.923076923\dots \text{ is sufficient.}$$

At $d = 5/4$, the condition is $D < 2$; the equality case $D = 2$ is not included. For $d > 5/4$, GIOW already gives the conclusion with no packing restriction [3]. The endpoints of all branches in (3) agree, and the packing inequality is strict throughout.

A new exact example is $d = 111/100 = 1.11$, $D = 49/40 = 1.225$. The adaptive profile cost is at most $644/5925$, giving the strict margin $31/23700$. The previous curve at this Hausdorff dimension was 1.22. An interpolated choice also covers $d = 57/50 = 1.14$, $D = 46/35$, with exact positive margin $214/1617375$. The proof supplies the exact parameter choices; the graph is illustrative and is not used to prove these inequalities.

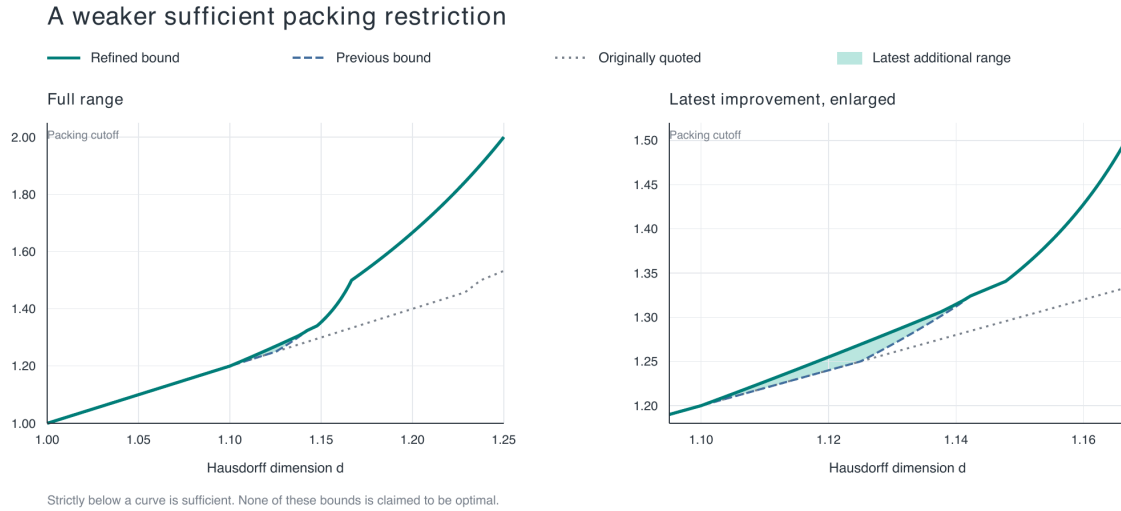


Figure 1: The refined sufficient curve, the preceding unforced curve, and the originally quoted curve. The second panel isolates the latest improvement. Values strictly below a curve are covered. No curve is asserted to be necessary.

1.1 Definitions and established inputs

A probability μ is s -Frostman if $\mu(B(x, r)) \leq Cr^s$ for $0 < r \leq 1$. Write

$$I_a(\mu) = \iint |x - x'|^{-a} d\mu(x) d\mu(x').$$

An s -Frostman probability of bounded support has finite a -energy whenever $0 < a < s$, by summing spatial annuli. The external inputs are as follows.

1. Frostman's lemma for Borel subsets of Euclidean space, and the countable bounded-cover characterization of packing dimension using upper box dimension. Closures preserve bounded upper box dimension; positive-mass compact restrictions are available by inner regularity.
2. Orponen's averaged radial-projection integrability [1], in the form of GIOW Theorem 3.7 or Keleti–Shmerkin Theorem 3.11. For separated compact Frostman probabilities of exponents greater than one, radial projections have L^q densities for some $q > 1$, with integrable q -th powers of their norms.

3. Plancherel, the Riesz energy identity, polar coordinates and their projection consequences, with Fourier normalization $\widehat{f}(\xi) = \int e^{-2\pi i x \cdot \xi} f(x) dx$.
4. Keleti–Shmerkin Lemma 3.4 and Corollary 3.5 [2], used only for finite terminal-cube restrictions, not to extract an infinite regular subset.
5. The elementary packet localization of GLOW Section 3 [3], Liu’s cap reconstruction (Lemma 4.2 of [5]), and the pinned quadratic identity [4]. The variable-chain inflation multiplier is derived below.

All constants may depend on the fixed compact supports, positive source-pin separation and chosen auxiliary parameters. They are uniform in frequency, regularized component, cube and cap. The notation $\text{RapDec}(R)$ means a quantity bounded by $C_M R^{-M}$ for every fixed $M > 0$, uniformly in those indices. Summing polynomially many such errors preserves this property.

2 Coherent positive linearizations and a conditional convergence theorem

This section proves the original branch of the bound, rather than assuming Proposition E.6 from the longer report. It uses positive affine approximations to distance and a summable first-norm comparison.

2.1 Angular regularity with a single exceptional set

Lemma 2.1 (Coupled-angle projection comparison). *Let $0 < \gamma < 1/2$, and let λ be a probability supported in $B(0, Cr)$, $0 < r \leq 1$, with $I_{1+2\gamma}(\lambda) < \infty$. For almost every $\theta \in S^1$, let $F(\theta) \in L^2(\mathbb{R})$ be the density of $(\pi_\theta)_*\lambda$, where $\pi_\theta(x) = x \cdot \theta$. There is a nonnegative function G with*

$$\|G\|_{L^2(S^1)}^2 \leq C_\gamma r^{2\gamma} I_{1+2\gamma}(\lambda)$$

such that, outside one angular null set, every pair satisfies

$$\|F(\theta) - F(\phi)\|_{L^2(\mathbb{R})} \leq C_\gamma d(\theta, \phi)^\gamma (G(\theta) + G(\phi)),$$

where d is circular arclength distance. In addition,

$$\int_{S^1} \sup_{|v| \leq Cr^2} \|F(\theta)(\cdot - v) - F(\theta)\|_2^2 d\theta \leq C_\gamma r^{4\gamma} I_{1+2\gamma}(\lambda).$$

Proof. We first prove the Hilbert-valued angular Sobolev estimate

$$\|F\|_{H^\gamma(S^1; L^2(\mathbb{R}))}^2 \leq C_\gamma r^{2\gamma} I_{1+2\gamma}(\lambda). \quad (4)$$

The projection-energy identity gives the projected densities and their joint second integrability, since finite $(1+2\gamma)$ -energy implies finite first energy on the bounded support. Fourier transform in the scalar projection variable. The transformed function is

$$f_\tau(\theta) = \widehat{\lambda}(\tau\theta).$$

Its angular derivative is a linear combination, with bounded angular coefficients, of $\widehat{\tau x_k \lambda}(\tau\theta)$, $k = 1, 2$. The Riesz energy identity for these signed measures gives

$$\int_{\mathbb{R}^2} |\widehat{x_k \lambda}(\xi)|^2 |\xi|^{2\gamma-1} d\xi \leq C_\gamma r^2 I_{1+2\gamma}(\lambda).$$

Indeed the spatial double integral has the factor $x_k x'_k$, whose absolute value is at most Cr^2 . Its absolute energy is finite, so approximation justifies the signed Fourier identity.

Angular Sobolev interpolation, or directly Holder on angular Fourier coefficients, gives

$$\|f_\tau\|_{H^\gamma}^2 \leq \|f_\tau\|_2^{2(1-\gamma)} \|f_\tau\|_{H^1}^{2\gamma}.$$

On $|\tau| \geq 1/r$, introduce the canceling weights

$$A_\tau = |\tau|^{2\gamma} \|f_\tau\|_2^2, \quad B_\tau = |\tau|^{-2(1-\gamma)} \|f_\tau\|_{H^1}^2.$$

Polar coordinates and the preceding signed-energy estimate show

$$\int_{|\tau| \geq 1/r} A_\tau d\tau \leq C_\gamma I_{1+2\gamma}(\lambda), \quad \int_{|\tau| \geq 1/r} B_\tau d\tau \leq C_\gamma r^2 I_{1+2\gamma}(\lambda).$$

For the undifferentiated part of B_τ , use $|\tau|^{-2} \leq r^2$; its derivative part has exactly the energy weight after the factor τ^2 is inserted. Holder in τ now bounds the integral of $\|f_\tau\|_{H^\gamma}^2$ by $C_\gamma r^{2\gamma} I_{1+2\gamma}(\lambda)$ on this region. When $|\tau| \leq 1/r$, both f_τ and its angular derivative are bounded uniformly, so the contribution is at most C/r . This is absorbed because

$$I_{1+2\gamma}(\lambda) \geq (Cr)^{-1-2\gamma}.$$

Plancherel in the scalar variable proves (4).

Choose a measurable Hilbert-valued representative and define

$$G(\theta)^2 = \int_{S^1} \frac{\|F(\theta) - F(z)\|_2^2}{d(\theta, z)^{1+2\gamma}} dz.$$

The angular Slobodeckij identity gives $\|G\|_2^2 \leq C_\gamma \|F\|_{H^\gamma}^2$. For completeness, expand in angular Fourier coefficients: translation invariance makes each nonzero coefficient contribute its squared Hilbert norm times

$$\int_{-\pi}^{\pi} \frac{|e^{ikh} - 1|^2}{|h|^{1+2\gamma}} dh \asymp_\gamma |k|^{2\gamma}.$$

Tonelli and Parseval prove the assertion also for Hilbert-valued functions.

Remove the null set where G is infinite, where F is undefined, or where it does not represent the projected density. Given two remaining angles at distance $\delta > 0$, take an arc J of length comparable to δ containing both, with every point of J within $C\delta$ of each endpoint. Average the triangle inequality through $z \in J$. Cauchy–Schwarz gives

$$\begin{aligned} \|F(\theta) - F(\phi)\|_2 &\leq |J|^{-1/2} \left[\left(\int_J \|F(\theta) - F(z)\|_2^2 dz \right)^{1/2} + \left(\int_J \|F(\phi) - F(z)\|_2^2 dz \right)^{1/2} \right] \\ &\leq C_\gamma \delta^\gamma (G(\theta) + G(\phi)). \end{aligned}$$

For large δ , the whole circle serves as J ; the diagonal case is immediate. This proves a single-null-set, all-pairs estimate, which can be evaluated on correlated angular laws.

Finally Plancherel bounds the translation expression, before angular integration, by

$$Cr^{4\gamma} \int_{\mathbb{R}} |\tau|^{2\gamma} |\widehat{\lambda}(\tau\theta)|^2 d\tau.$$

Here the supremum over $|v| \leq Cr^2$ was bounded inside the Fourier integral, using $|e^{-2\pi i\tau v} - 1|^2 \leq C_\gamma |\tau v|^{2\gamma}$. Polar coordinates give the required translation bound. $\square$

2.2 Consecutive positive linearizations

Let μ, ν be compact probabilities with positively separated supports, satisfying Frostman bounds with exponents $s, t > 1$. Choose

$$0 < \gamma < \min\left(\frac{1}{2}, \frac{s-1}{2}\right).$$

For a positive-mass dyadic source square Q , write $p_Q = \mu(Q)$, $\mu_Q = p_Q^{-1}\mu|_Q$. Orponen's theorem, in the GIOW Theorem 3.7 form stated in the preliminaries [1, 3], supplies $q > 1$ and

$$(\Theta_x)_*\nu = \rho_x d\sigma, \quad \Theta_x(y) = \frac{x-y}{|x-y|}, \quad B = \int \|\rho_x\|_q^q d\mu(x) < \infty,$$

where σ is normalized arclength. The hypotheses follow by choosing a common Frostman exponent between one and $\min(s, t)$. In every node choose $x_Q \in Q \cap \text{supp } \mu$ at which the density exists and

$$F_Q := \|\rho_{x_Q}\|_q^q \leq \frac{2}{p_Q} \int_Q \|\rho_x\|_q^q d\mu(x).$$

There are countably many nodes, and these choices give

$$\sum_{Q \text{ at level } n} p_Q F_Q \leq 2B \quad \text{at every level.}$$

Define the affine approximation

$$L_Q(x, y) = \Theta_{x_Q}(y) \cdot (x - y).$$

At level n , push the original positive pair probability by $(x, y) \mapsto (y, L_{Q_n(x)}(x, y))$. It has a nonnegative density f_n relative to $\nu(dy)dt$. To see this, each μ_Q has finite first energy and therefore second-integrable projected densities for almost every angle. Its chosen center's angular pin law is absolutely continuous, so the assertion holds for almost every pin, simultaneously at all countably many nodes. The conditional densities can be chosen jointly measurably. Each f_n has mass one, and Taylor's theorem gives

$$|L_{Q_n(x)}(x, y) - |x - y|| \leq C2^{-2n}$$

uniformly, once the squares are smaller than the fixed separation.

Proposition 2.2 (Coherent first-norm comparison). *For $r = 2^{-n}$ sufficiently small and every $A \geq 1$,*

$$\|f_{n+1} - f_n\|_{L^1(\nu \times dt)} \leq C \left[Ar^{1+4\gamma} \sum_{Q \text{ at level } n+1} p_Q I_{1+2\gamma}(\mu_Q) \right]^{1/2} + CBA^{1-q}. \quad (5)$$

Constants depend on the fixed support geometry and separation, the exponents and Frostman bounds, but not on n, A .

Proof. Let Q be a child of P , and apply both affine maps to the same conditional source μ_Q . Set $b = x_Q$, and recenter this source at b . Put $\theta = \Theta_{x_P}(y)$, $\phi = \Theta_b(y)$. Their separation is $O(r)$, and

$$L_P(x, y) = \theta \cdot (x - b) + \theta \cdot (b - y), \quad L_Q(x, y) = \phi \cdot (x - b) + |b - y|.$$

The constants differ quadratically:

$$|\theta \cdot (b - y) - |b - y|| = |b - y| |\theta \cdot \phi - 1| \leq Cr^2.$$

Recentring is essential here; estimating the uncentered translations separately would lose a first-order term.

Keep those pins for which both angular densities at θ and ϕ are at most A . The resulting joint angular law has both marginals bounded by $A\sigma$, while its deleted pin mass is at most

$$A^{1-q}(F_P + F_Q).$$

No independence of the angles or of angular and radial pin coordinates is used. Apply Lemma 2.1 to the translated conditional source. Its pointwise all-pairs bound is valid on the coupled angles, since both marginals avoid the single exceptional null set. Integrating the square and using the marginal bounds gives $CAr^{4\gamma} I_{1+2\gamma}(\mu_Q)$. The pin-dependent scalar translation has size

at most Cr^2 ; the lemma's translation estimate, with the relevant bounded angular marginal, gives the same bound. By the triangle inequality, the integrated squared second-norm difference of the two conditional affine densities is therefore at most

$$CAr^{4\gamma}I_{1+2\gamma}(\mu_Q).$$

For each pin both densities lie in a common interval of length Cr . Cauchy–Schwarz converts their second norm to their first norm with factor $Cr^{1/2}$. On deleted pins the first-norm difference of two probability densities is at most two. Decompose each parent conditional source into its children, sum with weights p_Q , and use Cauchy–Schwarz over those weights and retained pins. The good terms give the square root in (5). The bad terms sum to at most CBA^{1-q} , because

$$\sum_Q p_Q F_Q \leq 2B, \quad \sum_{Q \text{ at level } n+1} p_Q F_{\text{parent}(Q)} = \sum_{P \text{ at level } n} p_P F_P \leq 2B.$$

This proves the estimate. $\square$

2.3 The convergence criterion and its scope

Theorem 2.3 (Conditional-energy criterion). *Under the preceding assumptions, set*

$$Z_n = 2^{-n(1+4\gamma)} \sum_{Q \text{ at level } n+1} p_Q I_{1+2\gamma}(\mu_Q), \quad \eta_q = \frac{q-1}{2q-1}.$$

If $\sum_n Z_n^{\eta_q} < \infty$, then $(d_y)_ \mu$ is absolutely continuous for ν -almost every y .*

Proof. Enlarge B to at least one and omit finitely many scales so $Z_n \leq 1$. In Proposition 2.2 choose

$$A_n = (B/\sqrt{Z_n})^{1/(q-1/2)} \geq 1.$$

Both terms are bounded by $CB^{1/(2q-1)}Z_n^{\eta_q}$. Thus f_n is Cauchy in $L^1(\nu \times dt)$. Its nonnegative limit has mass one. Uniform approximation of the affine maps to distance identifies its weak measure limit with the pushforward of $\mu \times \nu$ by $(x, y) \mapsto (y, |x-y|)$. Uniqueness of disintegration then shows that, for almost every pin, the actual pinned law has the resulting density. In particular the conclusion is about the raw positive distance laws, not signed approximations. $\square$

2.4 The covering and Borel consequences

Proposition 2.4 (A covering-number sufficient condition). *Suppose additionally that the number of occupied source squares of side r is at most Cr^{-u} , uniformly for small dyadic r . If $u < 2s-1$, then the preceding criterion holds for a suitable γ . In particular an s -Ahlfors regular source, $s > 1$, satisfies this sufficient condition when the separated pin measure has a Frostman exponent greater than one.*

Proof. For $a < s$, integration of the source Frostman ball bound within a square of diameter $O(r)$ gives, for each x in that square,

$$\int_Q |x-x'|^{-a} d\mu(x') \leq Cr^{s-a}.$$

Integrate over $x \in Q$ and normalize twice. This proves

$$I_a(\mu_Q) \leq Cr^{s-a}/p_Q.$$

Taking $a = 1 + 2\gamma < s$ and summing over occupied child squares yields

$$\sum_Q p_Q I_{1+2\gamma}(\mu_Q) \leq C r^{s-1-2\gamma-u}, \quad Z_n \leq C r^{s-u+2\gamma}.$$

The strict assumption $u < 2s-1$ permits a choice $0 < \gamma < \min(1/2, (s-1)/2)$ with $s-u+2\gamma > 0$. The series then converges geometrically. An s -Ahlfors regular support has covering number $O(r^{-s})$, so one may take $u = s$. $\square$

For comparison, Keleti–Shmerkin [2, Theorem 1.2] obtain full Hausdorff dimension of pinned distance sets when the source packing dimension is at most twice its Hausdorff dimension minus one. The absolute-continuity assertion below is derived from the preceding positive-density comparison; the cited dimension theorem is not an absolute-continuity input.

Proposition 2.5 (Borel-set consequence). *Let $E, F \subset \mathbb{R}^2$ be Borel sets such that*

$$\dim_{\mathrm{H}} E > 1, \quad \dim_{\mathrm{P}} E < 2 \dim_{\mathrm{H}} E - 1, \quad \dim_{\mathrm{H}} F > 1.$$

Then $|\Delta_y(E)| > 0$ for some $y \in F$. More precisely, one can choose a compactly supported Frostman probability μ on a compact subset of E so that, for any prescribed compactly supported t -Frostman pin probability ν on F , $t > 1$, the law $(d_y)_\mu$ is absolutely continuous for ν -almost every pin. The source measure here is a selected restriction, not an assertion about an arbitrary original measure on E .*

Proof. Choose $1 < s < \dim_{\mathrm{H}} E$ and $u > \dim_{\mathrm{P}} E$ with $u < 2s - 1$. Frostman’s lemma gives an s -Frostman probability μ_0 compactly supported on E . Choose u_0 strictly between $\dim_{\mathrm{P}} E$ and u . The countable-cover characterization of packing dimension gives a cover by bounded sets of upper box dimension at most u_0 . Replacing these sets by their closures preserves upper box dimension and makes them compact. At least one member has positive μ_0 -mass. By inner regularity, choose a compact subset K of its intersection with E having positive μ_0 -mass. Set $\mu = \mu_0|_K / \mu_0(K)$. It is still s -Frostman, and its occupied-square count is $O(r^{-u})$, since $\dim_{\mathrm{B}} K \leq u_0 < u$.

It remains to remove the separation condition without changing this selected measure. Cover the complement of the diagonal in $\mathbb{R}^2 \times \mathbb{R}^2$ by countably many products of closed bounded rational balls $U \times V$ with positive separation. The interiors of such products already cover the complement. On each positive-mass product, normalize the restrictions of μ and the prescribed ν . These probabilities satisfy the Frostman and covering assumptions, have compact separated supports, and Proposition 2.4 applies. Their joint pinned-distance pushforward is absolutely continuous with respect to $\nu(dy)dt$.

The diagonal has $\mu \times \nu$ -mass zero because μ has no atoms. A set of zero $\nu \times dt$ -measure therefore has zero preimage mass on every member of the countable cover and hence under the full pair probability. Its full joint distance law is absolutely continuous. Disintegration proves the claimed almost-everywhere pinned assertion. Since these laws have mass one and are supported on $\Delta_y(K) \subset \Delta_y(E)$, their supports have positive length. Finally $\dim_{\mathrm{H}} F > 1$ supplies a compactly supported pin Frostman probability with exponent greater than one. $\square$

3 The finite-profile branch

For $1 < s \leq 5/4$ and $s \leq u \leq 2$, define

$$C(s, u) = U(s - 1, u - 1), \quad U(a, b) = \begin{cases} \frac{(b - a)(1 - a)}{2(2b - a)}, & a \leq b \leq 1/2, \\ \frac{b - a}{1 + 2b}, & 1/2 \leq b \leq 1. \end{cases} \quad (6)$$

The parameter a lies in $(0, 1/4]$, so all denominators are positive. The two expressions agree at $b = 1/2$.

Theorem 3.1 (Finite-profile criterion). *Let $E \subset \mathbb{R}^2$ be Borel, and write $d = \dim_{\mathbb{H}} E$, $D = \dim_{\mathbb{P}} E$. If*

$$1 < d \leq 5/4, \quad C(d, D) < d - 1, \quad (7)$$

then $|\Delta_y(E)| > 0$ for some $y \in E$.

3.1 The profile optimization

If g is a continuous function on $[0, N]$, put

$$c_g(m, n) = g(m) - \min_{m \leq x \leq n} g(x), \quad 0 \leq m < n \leq N.$$

The direction of this difference matters: the left endpoint is the larger physical cell. Call an edge $[m, n]$ admissible if $2n - m \leq N$.

Lemma 3.2 (A bounded-length chain). *Let N be a positive integer. Suppose $g(0) = 0$, g is 1-Lipschitz and piecewise linear on the integer grid, and*

$$ax \leq g(x) \leq bx, \quad 0 < a \leq b \leq 1, \quad b \geq \frac{1}{2}.$$

Given an integer $0 \leq n_0 < N$, there is an admissible decreasing chain from n_0 to zero, of at most

$$3 \left\lceil \log_2 \frac{N}{N - n_0} \right\rceil + 5$$

edges, with total cost at most $(bN - g(N))/(1 + 2b) + 3$. On a grid of mesh T , replace the additive constant by $3T$.

Proof. Define

$$L(x) = 2bx - bN - g(x), \quad V(x) = \frac{\max\{L(x), 0\}}{1 + 2b}.$$

The function L is nondecreasing. On $\{L \geq 0\}$, almost everywhere,

$$(-g')_+ \leq \frac{2b - g'}{1 + 2b}.$$

For $g' \geq 0$ this follows from $2b \geq 1$; for $g' < 0$ it is equivalent to $g' \geq -1$. Also

$$c_g(m, n) \leq \int_m^n (-g')_+ dx.$$

Let q be the largest integer with $L(q) \leq 0$. From $n_0 > q + 1$, repeatedly replace n by $\max\{q + 1, 2n - N\}$. These admissible edges double $N - n$, except possibly the last. Their total cost is

at most $V(n_0) - V(q + 1)$. If required, the edge from $q + 1$ to q costs at most one. If $n_0 \leq q$, omit this part.

In the region $L \leq 0$, for $n > N/2$ set $m_0 = 2n - N$. Then

$$g(m_0) \leq bm_0 \leq g(n).$$

Choose the leftmost grid minimum on $[m_0, n]$. It occurs before n , gives a zero-cost admissible edge, and remains in $L \leq 0$. If $n \leq N/2$, the edge to zero has zero cost since $g \geq 0$. Thus a finite zero-cost continuation exists. Merge consecutive edges greedily whenever their union is admissible. Their union still has zero cost. Maximality implies that every two merged edges more than double $N - n$, except at the end. This gives the stated edge count. Finally $V(n_0) \leq V(N) = (bN - g(N))/(1 + 2b)$. Rescaling proves the mesh version. $\square$

3.2 The unforced scale-profile estimate

An edge from n to $m < n$ in $[0, N]$ is admissible when $2n - m \leq N$, and its cost is

$$c_g(m, n) = g(m) - \min_{[m, n]} g.$$

No intermediate endpoint is prescribed in this section. This distinction is essential: the following estimate can be used in a distance argument only after the analytic transfer has been justified for such unforced chains.

Lemma 3.3 (Unforced profile bound). *Let $0 < a \leq b \leq 1$, and let $g : [0, N] \rightarrow \mathbb{R}$ be 1-Lipschitz with $ax \leq g(x) \leq bx$. For every $0 \leq n_0 < N$, an admissible decreasing chain from n_0 to zero exists with at most*

$$K \leq C \left(1 + \log \frac{N}{N - n_0} \right)$$

edges and total cost at most $NU(a, b)$, where

$$U(a, b) = \begin{cases} \frac{(b-a)(1-a)}{2(2b-a)}, & b \leq 1/2, \\ \frac{b-a}{1+2b}, & b \geq 1/2. \end{cases} \quad (8)$$

The formulas agree at $b = 1/2$. On a grid of mesh T containing $0, n_0, N$, the endpoints can be chosen on the grid with additional $O(KT)$ cost. If the barriers hold with uniform additive error ϵN , the additional cost is $O(K\epsilon N + KT)$.

Proof. The case $b \geq 1/2$ is the bounded-length unforced chain estimate of Lemma 3.2, followed by the grid limit described below. We prove the additional case $b \leq 1/2$, normalizing $N = 1$. Set

$$q = \frac{b}{2b-a}, \quad r = 2q - 1 = \frac{a}{2b-a}.$$

Then $1/2 < q \leq 1$ and $0 < r \leq 1$.

First observe that every endpoint $n \leq q$ admits free continuation to zero. For $1/2 < n \leq q$,

$$g(2n - 1) \leq b(2n - 1) \leq an \leq g(n), \quad (9)$$

because $(2b - a)n \leq b$. Choosing the leftmost minimum of g on $[2n - 1, n]$ gives an admissible zero-cost edge with endpoint less than n . When $n \leq 1/2$, the edge directly to zero is admissible and costs zero since $g(0) = 0$ and $g \geq 0$. On a finite grid the procedure terminates. Consecutive zero-cost intervals can be merged whenever their union is admissible, and their union remains

zero-cost. Every two consecutive unmergeable intervals more than double the complementary depth, so the retained zero-cost chain has the required logarithmic length. If $a = b$, then $q = 1$, and this argument already proves the result with zero cost. It also handles any starting point $n_0 \leq q$.

For the remaining initial portion, define on $[r, 1]$

$$H(z) = \frac{z}{2} + \left(b - \frac{1}{2}\right)r.$$

This majorizes bz , since

$$H(z) - bz = \left(\frac{1}{2} - b\right)(z - r) \geq 0.$$

For $x \geq q$, put

$$\Lambda(x) = H(2x - 1) - g(x) = x - \frac{1}{2} + \left(b - \frac{1}{2}\right)r - g(x).$$

The function Λ is nondecreasing, and

$$\Lambda(q) = H(r) - g(q) = br - g(q) = aq - g(q) \leq 0.$$

Almost everywhere,

$$(-g')_+ \leq \frac{1 - g'}{2} = \frac{\Lambda'}{2}.$$

Use maximal admissible downward jumps through $\{\Lambda > 0\}$, shortening the last jump if necessary to reach its lower boundary. Their costs sum to at most $\Lambda(n_0)_+/2$, because each edge cost is bounded by the negative variation of g over its interval. This is at most $\Lambda(1)/2$.

For a subsequent endpoint $n \geq q$ with $\Lambda(n) \leq 0$, the majorant is valid at $2n - 1 \geq r$, and gives

$$g(2n - 1) \leq H(2n - 1) \leq g(n).$$

Choose a leftmost minimum on $[2n - 1, n]$ to take a zero-cost edge. Continue until the endpoint becomes at most q ; it need not equal q . Below q , use (9) to continue for free. The maximal jumps in the positive region double complementary depth except at the last jump. Compressing the zero-cost intervals as above gives the claimed bound on the total number of edges.

Finally, $H(1) \geq g(1) \geq a$, and therefore

$$\frac{\Lambda(1)}{2} \leq \frac{1/2 + (b - 1/2)r - a}{2} = \frac{(b - a)(1 - a)}{2(2b - a)}.$$

This proves the cost bound.

For precision about arbitrary continuous profiles, first run the construction on piecewise affine interpolants on grids with mesh tending to zero. The interpolants retain both barriers and the Lipschitz constant. If n_0 is not a grid point, first jump to the nearest grid point below it; for sufficiently small mesh this is admissible and costs at most the mesh. Rounding the potential sign change and the threshold q requires at most a bounded number of grid crossings, each costing $O(T)$. For example, one stops the positive-potential construction at the first grid point above its lower boundary, then takes the adjacent grid edge across that boundary. Below q , the barrier argument applies directly. The edge-count bound is independent of the mesh. Pad endpoint lists by repeated endpoints and extract a convergent subsequence. Interval minima vary continuously with the profile and endpoints, and admissibility is closed. Removing repeated endpoints gives the exact continuous statement. This also proves the asserted fixed-mesh error bound.

For approximate barriers, clamp the profile between ax and bx and interpolate its grid values. This preserves the 1-Lipschitz property and changes the profile uniformly by $O(\epsilon N + T)$. Every edge cost changes by at most twice this amount. The resulting total error is $O(K\epsilon N + KT)$, as claimed. $\square$

The improvement over the global slope-1/2 majorant is quantified by

$$\frac{1/2 - a}{2} - U(a, b) = \frac{a(1/2 - b)}{2(2b - a)} \geq 0 \quad (b \leq 1/2).$$

The gain is strict below $b = 1/2$. This estimate is not asserted to be optimal in the small-slope range.

3.3 Finite regularization of the original pin measure

The constants in losses $R^{C_K h}$ below are independent of the regularization block size T ; fixed annular ratios depending on T change only multiplicative constants.

We prove a measure version of Theorem 3.1. Let μ, ν be probabilities on separated compact sets, normalized by a common translation and dilation into the interior of one unit dyadic cube. This preserves positive source-pin separation and both dimension exponents. Suppose both are t -Frostman for some $t > s > 1$, and

$$N(\text{supp } \nu, r) \leq Cr^{-u}, \quad 0 < r \leq 1. \quad (10)$$

We will prove

$$C(s, u) < s - 1 \implies (d_y)_* \mu \ll \mathcal{L}^1 \quad \text{for } \nu\text{-almost every } y. \quad (11)$$

Constants can depend on these fixed measures and their separation. The radial-projection theorem of Orponen, in the form of GIOW Theorem 3.7 or Keleti–Shmerkin Theorem 3.11, supplies $q > 1$ such that

$$B = \int \|\pi_*^y \mu\|_{L^q(S^1)}^q d\nu(y) < \infty. \quad (12)$$

Here $\pi^y(x) = (x - y)/|x - y|$. We also have $I_s(\mu) < \infty$.

Fix a large integer T . Use a smooth partition of frequency space into a low-frequency ball and annuli indexed by $R = 2^N$, $N = 4T\ell$. The annular ratios depend only on T ; all localization constants below may do so as well. Write P_R for the annular multiplier, including a fixed smooth spatial cutoff equal to one near $\text{supp } \mu$ and supported a positive distance from $\text{supp } \nu$. These multipliers have uniformly bounded convolution-kernel first norms, with constants depending on T .

Apply Keleti–Shmerkin’s finite regularization, Lemma 3.4 and Corollary 3.5, at depth N . More precisely, first distribute each depth- N cube mass uniformly within that cube. The lemma partitions all but mass $2^{-\epsilon N}$ into disjoint unions X_i of terminal cubes. Retain the *original* restrictions

$$w_i = \nu(X_i), \quad \sigma_i = w_i^{-1} \nu|_{X_i}.$$

They have the same coarser cube masses as the regularized measures. With

$$\Gamma = \epsilon + \frac{\log_2(4T + 2) + 2}{T},$$

we have $w_i \geq R^{-\Gamma}$. For each i , a function f_i , linear on the T -grid with slopes in $[0, 2]$, satisfies $f_i(0) = 0$ and, for every occupied cube of block depth k ,

$$2^{-f_i(k) - N/T} \leq \sigma_i(Q) \leq 2^{-f_i(k)}. \quad (13)$$

This follows directly by multiplying the one-step mass ratios in the definition of a regular measure. Also

$$sk - CTN - C_T \leq f_i(k) \leq uk + C_T. \quad (14)$$

For the first inequality use $\sigma_i \leq R^\Gamma \nu$, the Frostman bound, and the lower estimate in (13). For the second, the upper estimate in (13) forces at least $2^{f_i(k)}$ occupied cubes, whereas (10) bounds their number by $C2^{u_k}$. Interpolation extends the barriers with $O(T)$ error. Finally,

$$\int \|\pi_*^y \mu\|_q^q d\sigma_i(y) \leq BR^\Gamma. \quad (15)$$

Put $g_i = f_i - \text{id}$. Apply Lemma 3.3 with $a = s - 1$, $b = u - 1$. Start at the T -grid point immediately below $(1 - \zeta)N$, where $0 < \zeta < 1/8$ is fixed. We obtain a chain, common to the entire component,

$$n_0 > n_1 > \cdots > n_K = 0, \quad 2n_j - n_{j+1} \leq N, \quad (16)$$

$$K \leq C \log(1/\zeta) + C, \quad \sum_{j < K} c_{i,j} \leq C(s, u)N + CKTN + C_{T,K}, \quad (17)$$

$$c_{i,j} = g_i(n_{j+1}) - \min_{[n_{j+1}, n_j]} g_i.$$

No intermediate endpoint is prescribed. Admissibility of the final edge implies $n_{K-1} \leq N/2$, which is all that the final Fourier estimate will need. Grid interpolation in the clamping step only changes the last constant. The source μ is never restricted or regularized in this process.

3.4 Truncated conditional energies and deletion

For one edge write $m = n_{j+1}$, $n = n_j$, $b = 2^{-m}$, $a = 2^{-n}$, and $\rho = a/b$. A cube is called active when its σ_i -mass is positive. Only active cubes and their ancestors occur below.

Let Q be an active depth- m cube. For any concentric enlargement Q^+ by a factor between one and $R^{C_K h}$, set $\lambda = (\sigma_i)_{Q^+}$. For $m \leq k \leq n$, covering a ball by boundedly many dyadic cubes and using (13) gives

$$\lambda(B(z, 2^{-k})) \leq CR^{CT} 2^{f_i(m) - f_i(k)}.$$

Indeed, the denominator is at least the mass of the original active cube Q ; this avoids any boundary-sliver assumption about Q^+ . For intermediate radii between consecutive block-grid points, use the mass bound at the coarser grid point. The ratio between adjacent radii is at most 2^T , so this changes the estimate by a multiplicative constant depending only on T (one may take $O(2^{2T})$). It introduces no additional power of R . Summing dyadic annuli yields

$$J_\lambda(\rho) := \iint \min\{\rho^{-1}, b/|y - z|\} d\lambda(y) d\lambda(z) \leq C(N + 1)R^{CT} 2^{c_{i,j}}. \quad (18)$$

For separations larger than b , use the bound one. For the other annuli the summand is at most

$$CR^{CT} 2^{k - m + f_i(m) - f_i(k)} = CR^{CT} 2^{g_i(m) - g_i(k)}.$$

The innermost ball satisfies the same estimate. Replacing b , the tube width, or the truncation parameter by factors at most $R^{C_K h}$ changes (18) by at most another factor $R^{C_K h}$. This follows directly from the minimum kernel; no infinite conditional energy is used.

Here is the deletion argument with its multiplier exposed. If a probability λ is in a ball of radius b , take a ρ -net of unoriented directions and, in each direction, a bounded-overlap cover by ρb -wide tubes. Pair counting gives

$$\sum_T \lambda(T)^2 \leq C J_\lambda(\rho). \quad (19)$$

For fixed y, z , at most $C \min\{\rho^{-1}, b/|y-z|\}$ directions can place both points in such a tube; the spatial tube multiplicity in one direction is bounded. Integrating this statement proves (19). Enlarged supports and tubes introduce only powers of the enlargement factor.

For each pin, delete source directions whose angular maximal density exceeds a threshold L . The maximal inequality on the circle and Holder's inequality imply

$$(\mu \times \sigma_i)(\text{source-heavy pairs}) \leq C_q L^{1-q} B R^\Gamma. \quad (20)$$

For completeness, if v_y is the angular density, then $\int_{\{M v_y > L\}} v_y \leq C_q L^{1-q} \|v_y\|_q^q$: bound the length of that set by $C_q L^{-q} \|v_y\|_q^q$, then use Holder. Intervals centered at a retained direction therefore have mass at most a constant times their length times L .

Now declare a pin tube heavy when $\lambda(T) > H\rho$. Associate to it the unit-length source tube in the same direction, widened to angular width $C\rho$. If this tube contains any pair not source-heavy, its source mass is at most $CL\rho$. The fixed source-pin separation justifies this correspondence. Consequently the retained mass of pairs using heavy pin tubes is at most

$$CL\rho \sum_{\lambda(T) > H\rho} \lambda(T) \leq \frac{CL}{H} \sum_T \lambda(T)^2 \leq \frac{CL}{H} J_\lambda(\rho). \quad (21)$$

One can discretize all tube centers and directions here: replacing a tube by a containing member of a grid changes widths by fixed factors. Directions within $O(\rho)$ have the same containing tube after this enlargement. Thus the estimate also bounds the union over the finer angular caps used below, without paying their number.

Sum (21) over active parents with weights $\sigma_i(Q^+)$. Concentric enlarged cubes at a fixed scale have overlap at most the square of their dilation factor. Equations (18)–(21) therefore remain valid with an additional $R^{C_K h}$. Choose

$$L = R^{D_1 h}, \quad H_{i,j} = R^{D_2 h + C\Gamma} 2^{c_{i,j}}. \quad (22)$$

Here D_1 is sufficiently large in terms of q, K , and D_2 is sufficiently large compared with D_1, K . Then, for Γ sufficiently small compared with h , the deleted pair mass is at most $CR^{-c_1 h}$. The constant c_1 can be chosen larger than any fixed geometric overlap exponent needed below, by increasing D_1, D_2 . If $H_{i,j}\rho \geq 1$, there are no heavy pin tubes. Factors caused by an enlargement are covered by increasing D_1, D_2 ; their dependence on K, q is permitted.

3.5 Fourier iteration with no prescribed midpoint

We use the regularized component σ_i , the conditional energy estimate (18), and the deletion estimates (20)–(21). The chain required in this subsection has the following properties:

$$\begin{aligned} n_0 > n_1 > \cdots > n_K = 0, \quad 2n_j - n_{j+1} \leq N, \quad n_0 = (1 - \zeta)N + O(T), \\ K \leq K_0, \quad \sum_{j < K} c_{i,j} \leq \mathcal{C}N + CK\Gamma N + C_{T,K}, \quad c_{i,j} = g_i(n_{j+1}) - \min_{[n_{j+1}, n_j]} g_i. \end{aligned} \quad (23)$$

Here $0 < \zeta < 1/8$, K_0 is fixed independently of $R = 2^N$, and the depths lie on the T -grid. The exponent $\mathcal{C}$ is any bound supplied by a profile argument with the stated edge count and perturbation errors. For the original finite-profile criterion it is $C(s, u)$ in (6); for the minimum refinement it is $V(s-1, u-1)$ in (38). No depth is required to equal $N/2$. We retain $N \in 4T\mathbb{N}$, so the standard angular scale used below is dyadic.

Put

$$a_j = 2^{-n_j}, \quad r_j = Ra_j, \quad \delta_j = r_j^{-1}, \quad \rho_j = \frac{a_j}{a_{j+1}}, \quad \delta_* = R^{-1/2}.$$

The two elementary inequalities

$$\delta_j \leq \rho_j, \quad \delta_* \leq \rho_j \quad (0 \leq j < K) \quad (24)$$

have different roles. The first is equivalent to $r_{j+1} \leq r_j^2$. For the second, admissibility and $n_{j+1} \geq 0$ give

$$n_j - n_{j+1} \leq \min\{n_j, N - n_j\} \leq N/2.$$

The last edge also gives $n_{K-1} \leq N/2$, hence $\delta_{K-1} \leq \delta_*$. The second inequality in (24) permits the standard source wave packets to be retained even when the auxiliary Fourier caps become narrower than δ_* .

A fixed source packet decomposition and a separate label tree

Fix a smooth annular multiplier at the scale R . First partition it into the usual smooth standard-cap multipliers $\psi_{R,S}$, with angular width δ_* . The label S has a nominal dyadic angular arc of that length; its multiplier is supported in a fixed enlargement of the arc, and these supports have bounded overlap. These standard multipliers are fixed before any finer partition is made.

Use their usual source wave packets, with spatial tubes of width $R^{-1/2+h}$ and fixed bounded length. Write

$$M_T \mu = \chi \eta_T (\psi_{R,S(T)} \widehat{\mu})^\vee.$$

Here χ is a fixed smooth compact source cutoff, equal to one on a neighborhood of $\text{supp } \mu$, and with support separated from the pin set. For each standard label the smooth spatial functions η_T form a partition of unity on $\text{supp } \chi$. Thus summing all standard packets gives precisely $P_R \mu$, with the source cutoff convention used above. All packet localization estimates are applied only to these smooth standard multipliers and these standard spatial tubes.

We now define a separate angular tree for Fourier estimates. For each level with $\delta_j \geq \delta_*$, its labels group whole standard labels according to their nominal dyadic ancestors of width δ_j . Their multipliers are sums of the corresponding $\psi_{R,S}$. For $\delta_j < \delta_*$, a level label is a pair (S, I) , where I is a dyadic angular interval of length δ_j meeting the support of $\psi_{R,S}$; its multiplier is

$$\psi_{R,S}(\xi) \mathbf{1}_I(\xi/|\xi|).$$

The intervals partition the entire angular support of the standard multiplier, including the part outside its nominal arc. Consequently each such interval is within angular distance $O(\delta_*)$ of the standard label's direction. Use half-open angular intervals to specify the partition at endpoints. The endpoint choices do not affect any circular integral.

After the standard scale, parents keep the same label S and group the finer dyadic intervals. Before it, parents group whole standard labels. If an edge of the spatial chain crosses the standard angular scale, insert that scale in the angular tree only. This creates no additional spatial level and no additional inflation factor. The notation $\theta \prec \alpha$ below means descent in this label tree. It is a relation between labels; overlapping smooth standard supports cause no ambiguity in this relation.

Every level- j multiplier has angular support of width $O(\delta_j)$, and the supports have bounded overlap at each level. For fine labels this follows because at each direction only boundedly many standard multipliers are nonzero, and the dyadic intervals partition that direction separately for each such multiplier. For coarse labels it follows by grouping adjacent standard labels. No differentiability of a fine angular indicator will be used: these indicators partition spectral measures on circles, not source packets.

Coarsened marks and their bad-pair estimate

For a level- j Fourier label θ , define its effective label to be θ itself if $\delta_j \geq \delta_*$, and its standard parent S otherwise. Its angular width is

$$\bar{\delta}_j = \max\{\delta_j, \delta_*\} \leq \rho_j.$$

For each active a_j -cube $Q_j \subset Q_{j+1}$, mark an effective label bad when the enlarged a_j -by- a_{j+1} test tube, centered at Q_j and directed along that effective label, has mass greater than $H_{i,j}\rho_j$ under $(\sigma_i)_{Q_{j+1}^+}$. All Fourier labels with the same effective label receive the same mark. Descendant spatial cubes inherit the mark from their unique level- j ancestor. Use the thresholds (22).

For definiteness, the averaging cubes and test dilations may be chosen as

$$Q_j^+ = R^{(2j+2)h}Q_j, \quad \text{test-tube dilation } R^{(4K_0+20)h}.$$

For large R , child averaging cubes and their further R^h localizations fit inside the corresponding enlarged parent. Additional fixed geometric factors are absorbed into these dilations or their constants. The same enlarged parent is used in the marking condition, the truncated conditional energy, and the Fourier iteration.

These marks are unions of whole standard source-cap labels. At coarse levels this is true by their definition as groups of standard labels; at fine levels all descendants of S share the mark of S . For an initial pin cube Q_0 , remove a standard source packet when its enlarged tube $2T$ meets Q_0 and its standard label has any inherited bad mark. Call the retained source function G_{R,i,Q_0} .

The deletion estimates already proved apply to these marks. Indeed, directions within $O(\bar{\delta}_j)$ of an effective direction are within $O(\rho_j)$, so the same ρ_j -net of directions and the same containing tube grid cover the marked tubes. Their widths and conditional mass normalization are exactly those used in (18)–(21), up to the permitted enlargements. Source-heavy directions remain only an auxiliary device for estimating the bad mass; they are not additional Fourier marks.

The standard packet estimates give $\|M_T\mu\|_1 \lesssim \mu(CT) + \text{RapDec}(R)$, and rapidly small pinned first norm outside an enlarged T . A separated source-pin pair lies in only $R^{O(h)}$ enlarged standard packet tubes: their directions are within $O(R^{-1/2+h})$ of the connecting line, and the standard directions have spacing comparable to $R^{-1/2}$. Spatial tube multiplicity for a fixed direction is bounded. When a removed packet contributes to such a pair, its standard label and its effective bad label differ in direction by $O(\bar{\delta}_j)$; its packet uncertainty is $O(R^h\delta_*)$. By (24), both fit inside the enlarged conditional test tube based on that pair.

It follows, with $\text{Bad}_{R,i}$ the union of bad pairs estimated by the deletion argument, that

$$\int \|(d_y)_*(P_R\mu - G_{R,i,Q_0(y)})\|_1 d\sigma_i(y) \leq CR^{C_{K_0}h}(\mu \times \sigma_i)(\text{Bad}_{R,i}) + \text{RapDec}(R). \quad (25)$$

Only standard packets were summed here. There is no multiplicity equal to the number of auxiliary fine Fourier labels in a standard cap. Taking D_1, D_2 sufficiently large in terms of q, K_0 , and then Γ sufficiently small compared with h , makes the right side at most CR^{-c} for some $c > 0$, uniformly in the component.

For later reference, put $G_{R,y} = G_{R,i,Q_0(y)}$ on the component X_i , and zero on the discarded pin remainder. Multiplying by the component weights and summing proves

$$\int \|(d_y)_*(P_R\mu - G_{R,y})\|_1 d\nu(y) \leq CR^{-c'} \quad (c' > 0). \quad (26)$$

On the discarded remainder use $\|P_R\mu\|_1 \leq C_T$ and its mass at most $R^{-\epsilon}$. This uses the first norm of the whole annular multiplier, rather than a sum of absolute packet norms there.

Initial reconstruction and the exact inherited identity

Fix a circle radius $r \asymp_T R$, and let ω_r be its normalized arclength measure. Choose a nonnegative compactly supported smooth frequency function ψ whose inverse transform is bounded away from zero on the fixed pin region. For every terminal Fourier label ℓ , with multiplier $\psi_{R,\ell}$ specified by the tree above, set

$$F_\ell = ((\psi_{R,\ell} \widehat{\mu} d\omega_r) * \psi)^\vee.$$

The terminal label is a standard label if its width equals δ_* , and a pair (S, I) otherwise. The fixed frequency convolution makes these functions smooth, even when the fine angular partition uses indicators. Define coarser functions by summing terminal descendants. In particular, summing all terminal descendants of S gives the original standard-cap function exactly.

On Q_0 , the circular extension of the retained standard packets, multiplied by $\check{\psi}$, agrees up to a rapid error with the sum of the retained full standard-cap functions. To verify this statement, sum all packets of each standard label first. A packet with $2T \cap Q_0 = \emptyset$ has rapidly small circular extension on Q_0 , by the standard integration-by-parts packet estimate used in Liu's Lemma 4.2. Therefore retaining those remote packets from a bad standard label changes the extension there by only a rapid error.

The fixed source cutoff does not change this conclusion. The kernel of a smooth standard annular-cap multiplier is concentrated at transverse scale $R^{-1/2}$ and longitudinal scale R^{-1} . Outside a fixed neighborhood of $\text{supp } \mu$, that multiplier applied to μ , together with any prescribed finite number of derivatives, is rapidly small in integrable norms. Consequently replacing $\chi(\psi_{R,S} \widehat{\mu})^\vee$ by $(\psi_{R,S} \widehat{\mu})^\vee$ changes its circular extension by a rapid error, uniformly for the radii in question. All source functions used for pinned densities still include χ , preserving their fixed compact support and source-pin separation.

All terminal descendants of a standard label have the same survival decision at Q_0 . Thus, only after this whole-standard-cap reconstruction, partition each retained standard spectral measure into its terminal Fourier descendants. It follows that

$$\check{\psi}(y)(G_{R,i,Q_0} * \widehat{\omega}_r)(y) = F_{Q_0,\text{root}}(y) + \text{RapDec}(R), \quad y \in Q_0, \quad (27)$$

where the function on the right is the sum of surviving terminal labels. No fine angular cutoff has been applied to an individual source packet. In particular, no localization assertion about the spatial tails of such an individual fine piece is needed.

More generally, for an active level- j cube and a parent angular label α , define $F_{Q_j,\alpha}$ to be the sum of terminal descendants of α with no inherited bad mark at levels $j, j+1, \dots, K-1$. Level zero has a single parent label, the root. At level K , no marks remain, so $F_{Q_K,\ell} = F_\ell$. The level- j mark is constant on each level- j Fourier label; future marks are inherited from the parent cube. Hence

$$F_{Q_j,\alpha} = \sum_{\substack{\theta \prec \alpha \text{ at level } j \\ \theta \text{ good for } Q_j}} F_{Q_{j+1},\theta}. \quad (28)$$

This is an exact partition of terminal labels. The function on the right for a fixed parent and label is independent of the child cube. After the standard angular scale, every future mark depends only on the standard parent S ; there a fixed fine function is simply retained or omitted as a whole. This is compatible with the exact identity.

One threshold factor per inflation step

Let m_Q be normalized Lebesgue measure on Q , and define

$$\mathcal{E}_j = \sum_{Q_j \text{ active}} \sigma_i(Q_j^+) \sum_{\alpha \text{ at level } j-1} \int |F_{Q_j, \alpha}|^2 dm_{Q_j^+}.$$

The level-zero angular sum consists only of the root. Initial localization and (27) give

$$\int |G_{R, i, Q_0(y)} * \widehat{\omega}_r(y)|^2 d\sigma_i(y) \leq C_T R^{C\zeta + C_{K_0}h} \mathcal{E}_0 + \text{RapDec}(R).$$

Indeed each initial Fourier function has frequencies of size $O_T(R)$. Its pointwise square is controlled by a normalized average on an R^{-1} -ball with rapidly decreasing tails. Replacing that ball by an enlarged initial cube costs at most its area ratio $R^{2\zeta + O(T/N) + O_{K_0}(h)}$. Multiplication by its pin cube mass and summation give the displayed bound without any regularity assumption on the conditional pin measure.

We prove

$$\mathcal{E}_j \leq C_T R^{C_{K_0}h} H_{i,j} \mathcal{E}_{j+1} + \text{RapDec}(R). \quad (29)$$

Fix a parent cube $P = Q_{j+1}$ and a parent angular label. Proposition 9.1 below gives a full selected-subset Plancherel and weighted-kernel proof of the step; we describe its geometry here in the notation of this iteration. For its level- j descendants, the Fourier supports of the functions on the right of (28) lie in balls of radius $O_T(R/r_j)$ with bounded overlap. The fixed frequency smoothing preserves this assertion, since $R/r_j = 2^{n_j} \geq 1$. Local quadratic orthogonality on cubes of side $a_j = r_j/R$, followed by a cover of their enlargements, reduces the square in (28) to the sum of squares, with a loss $R^{O_{K_0}(h)}$ and a rapidly decreasing global-norm error.

For a fixed label θ , the support of $F_{P, \theta}$ lies in a rectangle of tangential width $O_T(R/r_j)$ and normal width $O_T(R/r_{j+1})$. The circular arc has normal width $O_T(R/r_j^2)$, at most $O_T(R/r_{j+1})$ by admissibility; the fixed frequency smoothing adds only bounded width, which is absorbed since $r_{j+1} \leq R$. Therefore local constancy applies on its dual tubes of dimensions a_j by a_{j+1} , oriented along the actual Fourier label.

Cover the parent by such tubes with bounded overlap. For any covering tube that meets good children, choose one of those good children. Its effective marking direction differs from the actual Fourier direction by $O(\bar{\delta}_j)$. In view of $\bar{\delta}_j \leq a_j/a_{j+1}$, the enlarged test tube of that child contains the covering tube and the enlarged children contributing to it, after the prescribed dilation factors. All these children lie in P^+ . Their enlargement multiplicity is at most a power $R^{C_{K_0}h}$. Thus their total weight is bounded by

$$R^{C_{K_0}h} H_{i,j} \frac{a_j}{a_{j+1}} \sigma_i(P^+).$$

This is exactly the conditional good-tube mass bound, with the same normalization as in deletion.

Local constancy bounds the squared supremum on a dual tube by its average on an R^h -enlargement, with a small-power loss and rapid tails. Relative to the parent average, the area ratio is $R^{O_{K_0}(h)} a_{j+1}/a_j$. It cancels the ratio in the last display. The tube cover and its enlarged localizations have multiplicity at most $R^{O_{K_0}(h)}$, and fit in P^+ under the choices of dilations above. Summing them gives the right side of (29) for this parent and cap. Adding omitted bad tubes on this nonnegative upper-bound side, and then summing all parent cubes and caps, proves (29).

At fixed K_0, T, h , the numbers of source packets, Fourier labels, and spatial cubes are polynomial in R . All relevant global Fourier norms also have polynomial bounds: the spectral densities

are bounded by a constant times $|\widehat{\mu}|$, the standard multipliers have bounded overlap, and the smoothing is fixed. The source functions are finite sums of standard packets with polynomial derivative bounds. One can therefore choose the rapidly decreasing localization errors to remain negligible after every sum in the argument. Hard fine angular partitions require no derivative bounds for these estimates.

The terminal estimate and the chain cost

Iterating (29), using (23) and (22), gives the multiplicative loss

$$R^{C+C\zeta+C_{q,K_0}h+CK\Gamma}.$$

At level K , the spatial scale is one and the functions are the terminal Fourier functions F_ℓ with no remaining marks. Their global Lebesgue norms bound the remaining unit-scale averages, up to the allowed enlargement factors. By Plancherel and Cauchy–Schwarz,

$$\sum_\ell \|F_\ell\|_2^2 \leq C_T R^{-1} \int_{S^1} |\widehat{\mu}(r\omega)|^2 d\omega. \quad (30)$$

For clarity, the normalized mass of the circle in a fixed-radius frequency ball is $O_T(R^{-1})$. For each terminal label, apply Cauchy–Schwarz to the convolution defining F_ℓ , using this mass bound. Integrate in frequency and sum the squared multipliers; their bounded overlap proves (30). This reasoning is valid for all terminal widths $\delta_{K-1} \leq R^{-1/2}$, including strictly finer widths. It does not introduce their number as a factor.

We conclude

$$\int |G_{R,i,Q_0(y)} * \widehat{\omega}_r(y)|^2 d\sigma_i(y) \leq C_T R^{-1+C+C\zeta+C_{q,K_0}h+CK\Gamma} \int_{S^1} |\widehat{\mu}(r\omega)|^2 d\omega + \text{RapDec}(R). \quad (31)$$

Together with (26), this supplies the two shell estimates needed in the following summation argument. Its parameter choices may be made in the same order: first ζ , fixing K_0 ; then the threshold constants in terms of q, K_0 ; then h ; and finally T, ϵ , making Γ sufficiently small. Fixed constants depending on the resulting annular ratio are absorbed for sufficiently large R .

In particular the proved transfer uses the cost of the unforced chain (23). The source packet decomposition remains at the standard scale throughout; only the spectral functions used for quadratic inflation are refined past that scale.

3.6 Summation, absolute continuity, and the Borel reduction

Fix $\eta > 0$ with $C(s, u) + \eta < s - 1$. Choose ζ so that $C\zeta < \eta/4$, fixing a bound for K . Next choose the constants D_1, D_2 in (22). Choose h small enough that all $C_{q,K}h$ losses are below $\eta/4$, and finally choose T large and ϵ small so that $\Gamma \ll h$ and $CK\Gamma < \eta/4$. These choices are fixed independently of R . Logarithms and fixed T, K constants are absorbed for sufficiently large R .

For $y \in X_i$ set $G_{R,y} = G_{R,i,Q_0(y)}$, and on the discarded pin remainder set $G_{R,y} = 0$. Equation (25) and the parameter choices give

$$\int \|(d_y)_*(P_R\mu - G_{R,y})\|_1 d\nu(y) \leq CR^{-c}, \quad c > 0. \quad (32)$$

On the remainder, use the uniformly bounded first norm of the *whole annular multiplier*; push-forward decreases total variation, so its contribution is at most $C_T 2^{-\epsilon N}$. One must not estimate it by the sum of absolute values of all packets.

Liu's pinned quadratic identity [4] bounds the radial density's squared norm by a constant times

$$\int_0^\infty |G_{R,y} * \widehat{w}_r(y)|^2 r \, dr.$$

The source-pin distances lie in a fixed interval bounded away from zero, so its radial weights are equivalent to the unweighted first-dimensional density norm. The identity is applied separately at each pin and therefore permits the dependence of $G_{R,y}$ on y . Frequencies outside a fixed enlargement of the annulus contribute rapidly decreasing errors, by the packet and cutoff localization. Integrating (31) and using

$$\int_{|\xi| \gtrsim TR} |\widehat{\mu}(\xi)|^2 \, d\xi \leq C_T R^{2-s} I_s(\mu)$$

now gives

$$\int \|(d_y)_* G_{R,y}\|_2^2 \, d\nu(y) \leq C R^{1-s+C(s,u)+\eta} I_s(\mu). \quad (33)$$

To combine components, multiply their bounds by w_i and sum. The sets X_i are disjoint, and $\sum_i w_i \leq 1$; no factor counting the components appears.

The exponents in (32) and (33) imply convergence of the bad shell densities in $L^1(d\nu \, dt)$, and of the good shell densities in $L^2(d\nu \, dt)$. All source functions have the same fixed compact support, so the radial variable ranges over a bounded interval. Include the smooth low-frequency density. Their sum is an integrable joint density.

To identify it, finite shell sums reconstruct μ as a distribution. If $\Phi(y, t)$ is a smooth test function, then

$$x \mapsto \chi(x) \int \Phi(y, |x - y|) \, d\nu(y)$$

is a smooth compactly supported test function: the cutoff χ keeps source and pins separated. Distributional reconstruction, followed by the stated norm convergence, identifies the joint density with the measure induced by $(y, x) \mapsto (y, |x - y|)$ on $\nu \times \mu$. Uniqueness of disintegration proves (11). In particular these pinned probabilities are nonzero and supported on the corresponding compact distance sets.

Finally suppose (7) holds. By continuity and strictness choose

$$1 < s < t < d, \quad D \leq u \leq 2, \quad C(s, u) < s - 1,$$

where $u > D$ if $D < 2$, and $u = 2$ if $D = 2$. Take a compactly supported t -Frostman probability on E . When $D < 2$, the packing-dimension covering characterization covers E by countably many bounded pieces of upper box dimension less than u . Taking their closures and intersecting with E makes a Borel cover without changing the upper box bounds. At least one piece has positive Frostman mass. By inner regularity take a compact subset of positive mass there. When $D = 2$, the original compact support already has the universal covering bound with $u = 2$. In either case split the compact set into two separated compact subsets of positive mass. Their normalized restrictions μ, ν retain the t -Frostman estimate; the pin support satisfies (10). Equation (11) supplies the required pin in E . This proves Theorem 3.1.

Only the unforced cost bound is used at each regularized component. No midpoint or minimum-location condition remains in the measure theorem. The algebra leading to the explicit dimension curve follows in the next section.

4 Refinement using the actual minima

4.1 Parameters and statement

Throughout this section assume

$$\frac{1}{8} \leq a \leq \frac{1}{6}, \quad 2a \leq b \leq \frac{1}{2}. \quad (34)$$

Define

$$\begin{aligned} k &= 2b - a, & q &= \frac{b}{k}, & h &= \frac{1+q}{2}, \\ \lambda &= \frac{1-2b}{4b}, & L_0 &= \frac{1-2a}{4}, & E_1 &= \frac{1-q}{3}, \\ E_2 &= \frac{(b-a)(1+6b-4a)}{6k}, & & & A_E &= \frac{1-a-3b}{3}, \\ \sigma_E &= \frac{6b-1-2a}{3a}, & S_F &= \frac{1-a}{3} - h, & \sigma_F &= \frac{a+2}{3a}, \\ S_V &= \frac{1-a}{3} - \frac{(1-b)h}{2}, & \sigma_V &= \frac{1-a}{6a}. \end{aligned}$$

All three quantities $\sigma_E, \sigma_F, \sigma_V$ are strictly positive. For σ_E , use $6b-1-2a \geq 10a-1 \geq 1/4$. Set

$$J_E = \frac{\sigma_E L_0 + \lambda A_E}{\sigma_E + \lambda} = \frac{(1-a)(b-a)(6b-1)}{24b^2 - 14ab + 3a - 4b}, \quad (35)$$

$$J_F = \frac{\sigma_F L_0 + \lambda S_F}{\sigma_F + \lambda}, \quad (36)$$

$$J_V = \frac{\sigma_V L_0 + \lambda S_V}{\sigma_V + \lambda}. \quad (37)$$

In particular, all the denominators in these expressions are positive. The refined cost is the explicit continuous function

$$V(a, b) = \max\{\min(E_2, J_E), \min(J_F, J_V)\}. \quad (38)$$

Lemma 4.1 (Profile bound retaining two actual minima). *Assume (34). Let $g : [0, N] \rightarrow \mathbb{R}$ be 1-Lipschitz, with $ax \leq g(x) \leq bx$. If $7N/8 \leq n_0 < N$, there is an admissible decreasing chain from n_0 to zero, without a prescribed intermediate endpoint, whose cost is at most $NV(a, b)$. Its number of edges satisfies*

$$K \leq C \left(1 + \log \frac{N}{N - n_0} \right)$$

with an absolute constant C . On a grid of mesh T containing $0, n_0, N$, the additional cost is $O(KT)$. Additive barrier errors of size ϵN increase this to $O(K\epsilon N + KT)$.

The proof is elementary. It uses an affine potential, the location and value of a minimum on $[q, 1]$, and a reflection that raises successive new minima to the same height. We provide the construction, including its bounded-length and stability assertions.

4.2 Two reusable chain constructions

Normalize $N = 1$. Write $\text{Var}^-(g; I)$ for the negative variation of g on an interval I , in the direction of increasing coordinate. Every edge cost is at most this negative variation on its interval. Since a Lipschitz function is absolutely continuous,

$$\text{Var}^-(g; [v, w]) = \int_v^w (-g')_+ \leq \frac{w - v + g(v) - g(w)}{2}. \quad (39)$$

The following potential construction will be used twice. Suppose that $H(z) = cz + d$, where $c \geq 1/2$, majorizes the profile at all left endpoints being considered. Define

$$\Lambda(x) = H(2x - 1) - g(x).$$

Then Λ is nondecreasing, and almost everywhere

$$(-g')_+ \leq \frac{2c - g'}{2c + 1} = \frac{\Lambda'}{2c + 1}. \quad (40)$$

Indeed, for $g' \geq 0$ the inequality is immediate, while for $g' \leq 0$ it reduces to $g' \geq -1$. Through the region $\Lambda > 0$, use maximal admissible downward edges, shortening the last edge to the lower boundary of that region. The cost is at most $\Lambda(n_0)_+/(2c + 1)$, by (39) and (40). If $\Lambda(n) \leq 0$, then

$$g(2n - 1) \leq H(2n - 1) \leq g(n),$$

so moving to a leftmost minimum on $[2n - 1, n]$ gives a zero-cost edge. Each such endpoint is strictly smaller unless the desired terminal point has already been reached.

In particular, suppose that t is a minimum of g on $[t, 1]$, with $g(t) = m$, and put $e = g(1)$. The cone $H(z) = m + z - t$ majorizes g on $[t, 1]$. Its potential is nonpositive at $(1 + t)/2$. The preceding construction, with $c = 1$, followed by a free jump to t as soon as the current endpoint is at most $(1 + t)/2$, gives a chain to t , whenever $n_0 \geq t$, of cost at most

$$\frac{1 - t + m - e}{3}. \quad (41)$$

This quantity is nonnegative by the Lipschitz inequality. No linear upper barrier is needed for this cone construction. Zero-cost steps before its last jump stay in $[t, 1]$, since their left endpoints are at least t .

Here and below, the construction can first be performed for a piecewise affine profile. A chosen interval minimum is either a breakpoint or the left endpoint of the interval, except for constant pieces, where the leftmost convention again chooses a breakpoint or the left endpoint. There are consequently finitely many steps: steps that do not encounter a new breakpoint double the complementary depth $1 - n$. The general continuous case follows by the compactness argument at the end of the proof. This avoids assuming termination of an arbitrary infinite sequence of successive minima.

4.3 The expanded free interval

For $n \leq q$, a free continuation to zero is always possible. If $1/2 < n \leq q$, then

$$g(2n - 1) \leq b(2n - 1) \leq an \leq g(n).$$

Choose a leftmost interval minimum to obtain a zero-cost edge. When $n \leq 1/2$, jump directly to zero. This jump is free since $g(0) = 0$ and $g \geq 0$.

Choose a minimum point t of g on $[q, 1]$, and write

$$m = g(t), \quad e = g(1).$$

We shall retain the following relations, rather than immediately replacing the minimum by the lower barrier:

$$t \geq q, \quad at \leq m \leq bq, \quad m \leq e, \quad e \geq a. \quad (42)$$

Define

$$Q_m = \frac{1 + m/b}{2}.$$

Then $q \leq Q_m \leq h$. For $q < n \leq Q_m$,

$$g(2n-1) \leq b(2n-1) \leq m \leq g(n).$$

Thus the free interval actually extends to Q_m , by the same leftmost-minimum construction and the already proved free continuation below q .

On $z \geq m/b$, the affine function

$$H_m(z) = \frac{z}{2} + \left(b - \frac{1}{2}\right) \frac{m}{b}$$

majorizes bz . Its potential at Q_m is $m - g(Q_m) \leq 0$. Apply the potential construction with $c = 1/2$ down to the expanded free interval, then continue for free. This gives an always available chain of cost at most

$$L(m, e) = \frac{1-2e}{4} - \lambda m \leq L_0 - \lambda m. \quad (43)$$

The first expression is nonnegative: it is $(H_m(1) - g(1))/2$. If the initial point is already in the free interval, its zero cost satisfies the same estimate.

We will compare this chain with chains whose costs increase with m . For any $\sigma > 0$, $\lambda \geq 0$, and real A , the elementary intersection inequality is

$$\min\{L_0 - \lambda m, A + \sigma m\} \leq \frac{\sigma L_0 + \lambda A}{\sigma + \lambda} \quad (m \in \mathbb{R}). \quad (44)$$

To check it, split at $m = (L_0 - A)/(\sigma + \lambda)$. To the left, use the increasing line; to the right, use the decreasing line. The argument includes $\lambda = 0$.

4.4 An early minimum

Suppose first that $t \leq h$. In the parameter range (34), $q \leq 2/3$, so $h \leq 5/6 < 7/8 \leq n_0$. The cone construction (41) can therefore reach t . Choose a minimum point v on $[2t-1, q]$. This interval is nonempty because $t \leq h$. The edge $t \rightarrow v$ is admissible and has cost

$$(g(v) - m)_+ \leq (b(2t-1) - m)_+.$$

Indeed, $g \geq m$ on $[q, t]$, and $g(v)$ is the minimum on $[v, q]$. Below q we continue for free. This yields cost

$$\frac{1-t+m-e}{3} + (b(2t-1) - m)_+. \quad (45)$$

If the positive part vanishes, $m \leq e$ implies that this cost is at most $(1-t)/3 \leq E_1$. Otherwise it is

$$A = \frac{1-e-3b+(6b-1)t-2m}{3}.$$

Since $e \geq a$, $m \geq at$, and $t \leq h$,

$$A \leq \frac{1-a-3b+(6b-1-2a)h}{3} = E_2. \quad (46)$$

Alternatively $t \leq m/a$ and $6b-1 > 0$ give

$$A \leq A_E + \sigma_E m.$$

Comparison with (43), using (44), bounds the cheaper chain by J_E . Consequently the early case has cost at most

$$\max\{E_1, \min(E_2, J_E)\}. \quad (47)$$

4.5 A late minimum and reflection

Now suppose $t > h$. Choose a minimum point u on $[q, h]$, and put

$$w = g(u), \quad z = b(2u - 1).$$

Here $q \leq u \leq h$, $w \geq m$, and $2u - 1 \leq q$. We distinguish whether the terminal left interval is certainly free.

First suppose $w \leq z$. Define a nondecreasing function

$$r(x) = \begin{cases} 0, & x \leq u, \\ w - \min_{[u, x]} g, & x \geq u, \end{cases} \quad f(x) = g(x) + r(x).$$

The reflected function f is 1-Lipschitz. Here is a direct check. For $u \leq x < y$, if the running minimum does not decrease on $[x, y]$, then $r(x) = r(y)$, so the assertion follows for g . Otherwise a new minimum is attained at some $v \in [x, y]$, and

$$f(y) = g(y) + w - g(v), \quad f(x) \geq w.$$

It follows that $f(y) - f(x) \leq y - v \leq y - x$. The lower bound follows from $r(y) \geq r(x)$ and $g(y) - g(x) \geq -(y - x)$. Gluing at u handles intervals crossing u .

Moreover, $f \geq w = f(u)$ on $[q, 1]$ and $f(1) = e + w - m$. Apply the cone construction to f , reaching $u < n_0$ with cost at most

$$\frac{1 - u - e + m}{3}.$$

The function is unchanged to the left of u . A terminal jump to a minimum on $[2u - 1, q]$ has cost at most $(z - w)_+$, after which the original free continuation below q applies. The reflection need not satisfy the original linear upper barrier to the right of u : the cone uses only its Lipschitz property and its minimum, while the terminal and free portions use its unchanged left part.

For every edge $n \rightarrow v$, monotonicity of r gives

$$c_g(v, n) \leq c_f(v, n) + r(n) - r(v).$$

The extra terms telescope over the entire chain, to at most $r(n_0) \leq w - m$. We have therefore constructed a chain for g of cost at most

$$\frac{1 - u - e - 2m}{3} + \max(w, z). \tag{48}$$

In the present case $w \leq z$, this is

$$\frac{1 - e - 3b + (6b - 1)u - 2m}{3} \leq E_2,$$

using $e \geq a$, $u \leq h$, $m \geq at \geq ah$, and $6b - 1 \geq 0$. The floor chain also costs at most $L_0 - \lambda ah$. At $m = ah$, the increasing line $A_E + \sigma_E m$ has value E_2 . Consequently (44) implies

$$\min\{E_2, L_0 - \lambda ah\} \leq J_E.$$

The late case with $w \leq z$ is bounded by $\min(E_2, J_E)$.

4.6 A free terminal interval: length and variation

It remains to consider $t > h$ and $w \geq z$. If $t \leq n_0$, use the cone to reach t , then maximal admissible edges to reach h . The edge $h \rightarrow u$ is admissible since $u \geq q = 2h - 1$, and it is free since u minimizes g on $[q, h]$. The next edge to a minimum on $[2u - 1, q]$ is also free: its cost is at most $(z - w)_+ = 0$. Finish with the free continuation below q .

The middle chain from t to h has cost at most $t - h$. Combining with the cone yields

$$\frac{1 - t + m - e}{3} + t - h = \frac{1 - e - 3h + 2t + m}{3} \leq S_F + \sigma_F m,$$

where the last step uses $e \geq a$ and $t \leq m/a$. Comparison with the floor chain proves the bound J_F .

For the same middle chain, (39) gives the sharper alternative

$$\text{Var}^-(g; [h, t]) \leq \frac{t - h + g(h) - m}{2} \leq \frac{t - h + bh - m}{2}.$$

Its total cost with the cone is therefore at most

$$\frac{1 - e}{3} - \frac{(1 - b)h}{2} + \frac{t - m}{6} \leq S_V + \sigma_V m.$$

Comparison with the floor chain proves the bound J_V .

If $t > n_0$, omit the cone and start the middle chain at n_0 . Its length is at most $t - h$ and its negative variation is at most $\text{Var}^-(g; [h, t])$. The omitted cone cost $(1 - t + m - e)/3$ is nonnegative. Both displayed upper bounds thus remain valid. In all cases $n_0 > h$, as checked in the early case. The late case with $w \geq z$ is bounded by $\min(J_F, J_V)$.

4.7 Removing a redundant term and controlling the chain

The cases proved so far give cost at most

$$\max\{E_1, \min(E_2, J_E), \min(J_F, J_V)\}.$$

In fact E_1 is redundant. First,

$$\frac{E_2}{E_1} = \frac{1 + 6b - 4a}{2} \geq 1.$$

Second, the numerator of $L_0 - \lambda ah - E_1$, after multiplication by its positive denominator $24b(2b - a)$, is

$$(4 - 6a)b^2 + (6a^2 - 7a)b + 3a^2.$$

Writing $b = 2a + x$, $x \geq 0$, this is

$$a^2(5 - 12a) + 9a(1 - 2a)x + (4 - 6a)x^2 \geq 0.$$

At $m = ah$, both lines used to define J_E consequently have value at least E_1 . Their intersection value is at least their minimum at any specified point, by (44). Thus $J_E \geq E_1$, proving precisely the cost (38).

Each positive-potential or middle portion consists of maximal edges $n \rightarrow 2n - 1$, apart from a bounded number of terminal shortened edges. Such a maximal edge doubles $1 - n$. Consecutive zero-cost edges can be merged whenever their union is admissible; the union still has zero cost because its lower endpoint is a minimum on both constituent intervals. After all possible adjacent merges, any two consecutive retained edges, with endpoints $n_i > n_{i+1} > n_{i+2}$, satisfy

$$2n_i - n_{i+2} > 1, \quad \text{hence} \quad 1 - n_{i+2} > 2(1 - n_i).$$

Every construction above has only a bounded number of such portions and special terminal edges. It follows that its number of edges is at most $C(1 + \log(1/(1 - n_0)))$.

To justify the continuous assertion fully, interpolate the given profile on finer and finer partitions containing 0, n_0 , 1. Linear interpolation preserves both linear barriers and the Lipschitz constant. Run the finite piecewise affine constructions proved above. Their costs obey the same bound, and their lengths have the uniform bound just established. Pad endpoint lists by repeated endpoints and pass to a subsequence. The profiles converge uniformly. Minima on intervals vary continuously under uniform convergence and convergence of their endpoints. Admissibility is closed. The limiting chain therefore has the asserted cost and length; removing repeated endpoints gives a strictly decreasing chain.

For a prescribed grid of mesh T , round every endpoint down. If $m < n$, $m \geq 2n - N$, and N/T is integral, then

$$\left\lfloor \frac{m}{T} \right\rfloor \geq \left\lfloor \frac{2n - N}{T} \right\rfloor \geq 2 \left\lfloor \frac{n}{T} \right\rfloor - \frac{N}{T}.$$

Thus admissibility is preserved. An endpoint moves by at most T , and the minimum of a 1-Lipschitz function changes by at most T under this change of interval. The increase in cost is at most $2T$ per edge. Remove repetitions. Finally, clamping between ax and bx preserves the Lipschitz constant and changes a profile with approximate barriers uniformly by at most ϵN . Each edge cost changes by at most twice this amount. These arguments prove all stability and length claims in Lemma 4.1.

4.8 The exact cutoff is a unique root

Proposition 4.2. *For every $a \in [1/8, 1/6]$, there is a unique $b_*(a) \in [2a, 1/2]$ such that $V(a, b_*(a)) = a$. The function b_* is continuous,*

$$b_*(1/8) = 1/4, \quad b_*(1/6) = 1/2,$$

and $2a < b_*(a) < 1/2$ in the interior. For parameters (34),

$$V(a, b) < a \iff b < b_*(a).$$

Proof. We first show that $V(a, b)$ is strictly increasing in b , for fixed a . Both factors $(b - a)/(2b - a)$ and $1 + 6b - 4a$ are positive and strictly increasing, so E_2 is strictly increasing. Writing $D_E = 24b^2 - 14ab + 3a - 4b > 0$, differentiation gives

$$\frac{\partial J_E}{\partial b} = \frac{a(1 - a)(60b^2 - 12b + 1 - 4a)}{D_E^2} > 0.$$

For the last sign, the quadratic in b is increasing for $b \geq 2a \geq 1/4$; at $b = 2a$ it is $240a^2 - 28a + 1 \geq 5/4$, since its derivative in a is positive for $a \geq 1/8$.

For J_F, J_V , the corresponding σ is independent of b , $\lambda'(b) < 0$, and S_F, S_V are strictly increasing in b . Indeed,

$$q'(b) = -\frac{a}{(2b - a)^2} < 0, \quad h' = q'/2 < 0,$$

so $S'_F = -h' > 0$, and $S'_V = h/2 - (1 - b)h'/2 > 0$. Both S_F and S_V are less than L_0 . The former is negative; for the latter,

$$S_V - L_0 = \frac{1 + 2a}{12} - \frac{(1 - b)h}{2} \leq \frac{1}{9} - \frac{3}{16} < 0,$$

using $h \geq 3/4$. Thus the derivative of either intersection value $J = (\sigma L_0 + \lambda S)/(\sigma + \lambda)$ is

$$J' = \frac{\sigma \lambda' (S - L_0) + \lambda (\sigma + \lambda) S'}{(\sigma + \lambda)^2} > 0.$$

The minimum or maximum of finitely many strictly increasing functions is strictly increasing: compare all their values at two ordered arguments. This proves the strict monotonicity and continuity of V .

At $b = 2a$, direct substitution gives

$$E_2 = \frac{1 + 8a}{18} < a, \quad J_F = \frac{5 - 2a}{2(19 - 4a)}, \quad J_V = \frac{3 - 2a - 16a^2}{4(7 - 16a)}.$$

In the stated range $J_V \leq a$, with equality only at $a = 1/8$: the numerator of $J_V - a$ is $3(1 - 8a)(1 - 2a)$. At $a = 1/8$, $J_F > a$, so $V(1/8, 1/4) = 1/8$; in the interior $V(a, 2a) < a$. At $b = 1/2$,

$$J_E = J_F = J_V = L_0, \quad E_2 = \frac{1 - 2a}{3} > L_0,$$

and hence $V(a, 1/2) = (1 - 2a)/4 \geq a$, with equality only at $a = 1/6$. The intermediate value theorem and strict monotonicity give the unique root and the claimed endpoint and inequality statements.

For continuity, let $a_j \rightarrow a$ and take any convergent subsequence of $b_*(a_j)$, with limit b . The parameter constraints pass to the limit, and joint continuity gives $V(a, b) = a$. Uniqueness implies $b = b_*(a)$. Compactness then proves continuity of the full sequence, including at the endpoints. $\square$

4.9 Comparison, an exact example, and numerical values

The new bound improves the previous low-slope cost

$$U(a, b) = \frac{(b - a)(1 - a)}{2(2b - a)} = \frac{(1 - a)(1 - q)}{2}.$$

Here is an algebraic check that uses no numerical optimization. At $m = aq$, the decreasing floor line has value U ; the increasing early line has value $2U/3$, while the increasing free-terminal line has value

$$S_F + \sigma_F a q = -\frac{(1 + 2a)(1 - q)}{6} < 0.$$

Both intersections therefore occur to the right of aq . Their values satisfy $J_E, J_F \leq U$, strictly when $b < 1/2$ because then $\lambda > 0$. Consequently

$$V(a, b) \leq U(a, b), \quad V(a, b) < U(a, b) \quad \text{if } b < 1/2. \quad (49)$$

At $a = 3/20$, $b = 7/20$, exact arithmetic gives

$$E_2 = \frac{5}{33}, \quad J_E = \frac{187}{1255}, \quad J_F = \frac{1000}{6919}, \quad J_V = \frac{937}{6424}.$$

It follows by rational comparison that

$$V(3/20, 7/20) = \frac{187}{1255}, \quad \frac{3}{20} - V(3/20, 7/20) = \frac{1}{1004} > 0.$$

This is the exact profile margin used for the example in the main theorem. The cutoff at this dimension also has a simple exact expression:

$$b_*(3/20) = \frac{7 + \sqrt{13}}{30}, \quad B_M(23/20) = \frac{37 + \sqrt{13}}{30}. \quad (50)$$

To verify the root identification, put $b = (7 + \sqrt{13})/30$. Substitution gives $J_E = 3/20$ and $E_2 - 3/20 = (2\sqrt{13} - 7)/60 > 0$. Also $b < 9/25$, and strict monotonicity together with the exact calculation

$$J_F(3/20, 9/25) - 3/20 = -\frac{136}{51015} < 0$$

shows that $\min(J_F, J_V) < 3/20$. Thus $V(3/20, b) = 3/20$, and Proposition 4.2 identifies the claimed unique root.

For convenient evaluation, the level equations for E_2, J_E, J_F, J_V to equal a , after clearing positive denominators, are respectively

$$\begin{aligned} 0 &= 6b^2 + (1 - 22a)b + 10a^2 - a, \\ 0 &= (6 - 30a)b^2 + (20a^2 - a - 1)b + a - 4a^2, \\ 0 &= (8a^2 - 34a + 8)b^2 + (-4a^3 + 4a^2 - 9a)b + a^2 + 8a^3, \\ 0 &= -18ab^3 + (4 - 17a + 94a^2)b^2 - (3a + 19a^2 + 44a^3)b + 16a^3 - a^2. \end{aligned}$$

One can alternatively evaluate (38) directly and use monotone bisection. The following decimals are illustrative evaluations of the unique root; they play no role in the proof or in the definition of b_* .

a	$b_*(a)$, approximately	Expression attaining the root
0.125	0.250000000	J_V
0.130	0.269213180	J_V
0.135	0.289846951	J_V
0.140	0.312168690	J_V
0.142	0.322586562	J_F
0.145	0.332270658	E_2
0.150	0.353518376	J_E
0.155	0.386846972	J_E
0.160	0.427797338	J_E
0.165	0.479534898	J_E
1/6	0.500000000	$J_E = J_F = J_V$

Numerically, the active expressions change near $a = 0.140503259$, 0.142267709 , and 0.147793654 , in the order J_V, J_F, E_2, J_E . These approximate transition locations are not assertions of an additional piecewise algebraic theorem; the proved, continuous, unique-root definition already covers the full interval.

Remark 4.3 (Distinction from the usual total-drop orientation). Keleti–Shmerkin’s total-drop framework [2, Definition 5.1 and Proposition 5.2] also combines interval minima with dyadically related endpoints. Its admissible partitions have neighboring scales in ratio at most two, and its summands subtract an interval minimum from the value at the smaller scale. Reflecting our coordinate, $x = 1 - n$, turns our admissibility into $x_{i+1} \leq 2x_i$, but turns our cost into $f(x_{i+1}) - \min_{[x_i, x_{i+1}]} f$, which uses the larger endpoint. The affine barriers are also reflected. Thus that proposition cannot be substituted with the same parameters for the lemma proved here. Our proof establishes the required orientation directly; it does not claim an optimal profile estimate in the present low-slope range.

5 Endpoint normalization and adaptive intervals

5.1 An exact optimized cost

Throughout this section put

$$\frac{1}{10} \leq a \leq \frac{1}{6}, \quad 2a \leq b \leq \frac{1}{2}, \quad 0 \leq \theta \leq 1. \quad (51)$$

The parameter θ interpolates between a fixed head interval and an interval determined by the actual tail minimum. Define

$$\begin{aligned} k &= 2b - a, & q &= \frac{b}{k}, & h_\theta &= \frac{1+q}{2} - \frac{\theta a}{4k}, & r_\theta &= \frac{\theta}{4b}, \\ L_0 &= \frac{1-2a}{4}, & \lambda &= \frac{1-2b}{4b}, & A_0 &= \frac{1-a-3b}{3}, & \gamma &= \frac{6b-1-2a}{3a}. \end{aligned}$$

Here $\gamma \geq 0$, and $\gamma + \lambda > 0$, including the corner $a = 1/10, b = 1/5$, where $\gamma = 0$. Set

$$m_\theta = \frac{ah_\theta}{1-ar_\theta}, \quad E_\theta = A_0 + \gamma m_\theta, \quad J_E = \frac{\gamma L_0 + \lambda A_0}{\gamma + \lambda}, \quad (52)$$

$$F = \frac{(1-a)(b-a)}{4(2b-a)(1-b)}, \quad Z = \min \left\{ \frac{1-q}{3}, F \right\}. \quad (53)$$

For the two late routes, indexed by v (variation) and ℓ (length), put

$$\begin{aligned} A_v &= \frac{1-a}{3} - \frac{(1-b)h_\theta}{2}, & \sigma_v &= \frac{1-a}{6a} - \frac{(1-b)r_\theta}{2}, \\ A_\ell &= \frac{1-a}{3} - h_\theta, & \sigma_\ell &= \frac{a+2}{3a} - r_\theta, \\ P_i &= A_i + a\sigma_i, & J_i &= \frac{\sigma_i L_0 + \lambda A_i}{\sigma_i + \lambda} \quad (i = v, \ell). \end{aligned}$$

Both σ_i are strictly positive: for example $\sigma_v \geq (5-5a)/(48a) > 0$, using $b \geq 2a$. Also $1-ar_\theta \geq 7/8$. Thus all denominators are positive. Our new cost and its compact optimization are

$$C_\theta(a, b) = \max \{Z, \min(E_\theta, J_E), \min(P_v, J_v, P_\ell, J_\ell)\}, \quad (54)$$

$$C_A(a, b) = \min_{0 \leq \theta \leq 1} C_\theta(a, b). \quad (55)$$

These are exact real expressions. The last minimum exists by continuity on the compact interval; it is not an assumption about a numerical optimizer.

Lemma 5.1 (Normalized adaptive profile bound). *Under (51), let $g : [0, N] \rightarrow \mathbb{R}$ be 1-Lipschitz and $ax \leq g(x) \leq bx$. For $15N/16 \leq n_0 < N$, there is an admissible chain from n_0 to zero of cost at most $NC_\theta(a, b)$, and hence one of cost at most $NC_A(a, b)$. The number of edges is $O(1 + \log(N/(N-n_0)))$, uniformly in θ . Grid and approximate barrier errors are $O(KT + K\epsilon N)$, as in Lemma 4.1.*

5.2 Endpoint normalization and free continuation

Normalize $N = 1$. Replace $g(x)$ by $\tilde{g}(x) = \min\{g(x), 1+a-x\}$. This preserves the barriers and the Lipschitz constant, and sets $\tilde{g}(1) = a$. Every edge cost increases. Indeed, if its left endpoint v is unchanged, the interval minimum can only decrease. Otherwise

$$c_{\tilde{g}}(v, n) \geq (1+a-v) - (1+a-n) = n-v \geq c_g(v, n).$$

It is therefore enough to construct chains for a profile with $g(1) = a$.

Let t minimize g on $[q, 1]$, and write $m = g(t)$. Then

$$aq \leq at \leq m \leq a, \quad m \leq bq. \quad (56)$$

Every endpoint below q has a free continuation, by $g(2n-1) \leq b(2n-1) \leq an \leq g(n)$ for $1/2 < n \leq q$, and the free jump to zero below $1/2$. The free region extends to

$$Q_m = \frac{1 + m/b}{2},$$

because $g(2n-1) \leq m \leq g(n)$ when $q < n \leq Q_m$. Set

$$Q_\theta = (1 - \theta)q + \theta Q_m, \quad H_\theta = \frac{1 + Q_\theta}{2} = h_\theta + r_\theta m.$$

Thus every endpoint below Q_θ has a free continuation, $q \leq Q_\theta \leq Q_m$, and $H_\theta \leq (3 + q)/4 \leq 11/12 < n_0$.

The affine majorant $z/2 + (b - 1/2)m/b$, applied above Q_m , has nonpositive potential at Q_m . The potential construction (40) therefore supplies, independently of all later cases, a chain with cost

$$L(m) = L_0 - \lambda m. \quad (57)$$

5.3 Early minima and the supporting line

If $t \leq Q_\theta$, the cone chain to t followed by free continuation costs at most $(1 - t + m - a)/3$. If $Q_\theta < t \leq H_\theta$, jump after that cone to a minimum on $[2t - 1, Q_\theta]$, and then continue freely. The cost is at most

$$\frac{1 - t + m - a}{3} + (b(2t - 1) - m)_+. \quad (58)$$

When the terminal contribution is zero, the cone bound is at most $(1 - q)/3$. A second construction gives F . On $[t, 1]$, $g(z) \leq \min\{bz, m + z - t\}$. These two lines meet at $z_* = (t - m)/(1 - b)$. The affine function

$$H(z) = \frac{z}{2} + \left(b - \frac{1}{2}\right) z_*$$

majorizes their minimum. Its potential bounds the cost down to $(1 + t)/2$, followed by a free jump to t , by

$$\frac{H(1) - a}{2} = \frac{1/2 - a - \kappa(t - m)}{2}, \quad \kappa = \frac{1/2 - b}{1 - b} \geq 0.$$

Here the construction can stop with positive remaining potential: its discarded value is nonnegative. If the potential reaches zero first, use zero-cost minimum steps until the terminal jump is admissible. All left endpoints where the majorant is used are at least t . Since $t - m \geq q - a$, the displayed expression is at most F , by direct substitution. The zero-terminal case thus has cost at most Z .

If the terminal contribution in (58) is positive, the bound becomes

$$A(t, m) = \frac{1 - a - 3b + (6b - 1)t - 2m}{3}.$$

Using $t \leq m/a$ gives the increasing line $A_0 + \gamma m$. Using $t \leq H_\theta$ gives a decreasing line, whose slope is $((6b - 1)r_\theta - 2)/3 < 0$. They intersect at m_θ , with common value E_θ . Comparing the increasing line with the independent floor (57) also gives J_E . Hence this case has cost at most $\min(E_\theta, J_E)$.

5.4 Late minima: reflection and two terminal routes

Suppose $t > H_\theta$, and let u minimize g on $[Q_\theta, H_\theta]$. Put $w = g(u)$, $z = b(2u - 1)$. Then $w \geq m$, $2u - 1 \leq Q_\theta$, and $m \geq at > aH_\theta$, so $m > m_\theta$.

If $w \leq z$, use the running-minimum reflection from the preceding section: set $r = 0$ before u , $r(x) = w - \min_{[u,x]} g$ after u , and $f = g + r$. It is 1-Lipschitz, has minimum w on $[Q_\theta, 1]$, and $f(1) = a + w - m$. The edge inequality $c_g(v, n) \leq c_f(v, n) + r(n) - r(v)$ transfers a cone chain to u , its terminal jump, and free continuation, with total cost at most

$$\frac{1 - u - a - 2m}{3} + \max(w, z) = \frac{1 - a - 3b + (6b - 1)u - 2m}{3}.$$

Since $u \leq H_\theta < m/a$, both previous affine bounds apply; the decreasing one is evaluated to the right of m_θ . The bound is again $\min(E_\theta, J_E)$.

If $w \geq z$, the interval below H_θ is free: jump first to u , then to a minimum on $[2u - 1, Q_\theta]$, then continue freely. Above it, use the cone to t and an admissible chain from t to H_θ . Bounding this middle part by negative variation or by its length gives respectively

$$\frac{1 - a}{3} - \frac{(1 - b)H_\theta}{2} + \frac{t - m}{6}, \quad \frac{1 - a}{3} - H_\theta + \frac{2t + m}{3}.$$

The relation $t \leq m/a$ bounds these by $A_i + \sigma_i m$, $i = v, \ell$. If $t > n_0$, omit the cone and restrict the middle variation or length to $[H_\theta, n_0]$; the omitted cone allowance is nonnegative, so the bounds remain valid. Since $m \leq a$, each increasing line is at most P_i . Intersecting it with the decreasing floor gives J_i . The late free-terminal case therefore has cost at most $\min(P_v, J_v, P_\ell, J_\ell)$, completing the cost proof.

There are a bounded number of chain portions. For clarity, compression does not require that their edges be free: for $l < m < n$,

$$c_g(l, n) \leq c_g(l, m) + c_g(m, n).$$

This follows by comparing the two interval minima and using $\min_{[l,m]} g \leq g(m)$. Merge any adjacent pair whose union is admissible. In a chain with no further such merge, complementary depth more than doubles every two edges, giving the claimed logarithmic count. For piecewise affine profiles the constructions terminate; approximate a general Lipschitz profile by piecewise affine interpolation and pass to a convergent subsequence of uniformly bounded endpoint lists. Costs, minima and admissibility pass to the limit. Rounding down and clamping the barriers give the stated errors exactly as before. All bounds are uniform in θ ; this proves Lemma 5.1.

5.5 The unique root and the dimension curve

Lemma 5.2. *The function C_A is continuous and strictly increasing in b for fixed a in (51). There is a unique continuous $\beta(a) \in [2a, 1/2]$ satisfying $C_A(a, \beta(a)) = a$. Its endpoint values are $\beta(1/10) = 1/5$, $\beta(1/6) = 1/2$, and $\beta(a) > 2a$ for $1/10 < a \leq 1/6$.*

Proof. All expressions are jointly continuous with positive denominators; minimizing over a fixed compact interval preserves continuity. We check strict increase of every ingredient in (54). Both terms defining Z increase strictly. Write

$$E_\theta = \frac{b - a}{3(2b - a)} T, \quad T = \frac{12b^2 + (2 - 8a)b - \theta a(1 - a)}{4b - \theta a}.$$

The first factor increases strictly. We have $T > 0$, and its derivative has positive numerator $48b^2 - 24ab\theta + 2a\theta + 4a^2\theta$. Thus E_θ increases strictly. For J_E , its derivative is a positive factor times $60b^2 - 12b + 1 - 4a$. Substitute $b = 2a + x$, $x \geq 0$, to obtain

$$240a^2 - 28a + 1 + (240a - 12)x + 60x^2 > 0.$$

Indeed, the constant term is increasing for $a \geq 1/10$ and is $3/5$ at $a = 1/10$. Thus J_E increases strictly.

For the late bounds, regard $H_\theta(m) = h_\theta + r_\theta m$ as a function of b at fixed $a, \theta, m > 0$. Then

$$\partial_b H_\theta(m) = -\frac{a(1-\theta)}{2(2b-a)^2} - \frac{\theta m}{4b^2} < 0.$$

Both route lines $A_i + \sigma_i m$ therefore increase strictly in b , as does $L(m)$, since $\partial_b L(m) = m/(4b^2) > 0$. At $m = aq$, the differences between the floor and the two route lines are

$$\frac{(b-a)(5-2a-9b)}{12(2b-a)} > 0, \quad \frac{(4-a)(b-a)}{6(2b-a)} > 0.$$

Their intersections are consequently at positive m . To compare intersection values at two upper slopes, evaluate the two new lines at the old intersection: both values have increased strictly. The maximum of the minimum of a decreasing and an increasing affine line is their intersection value, so J_i increases strictly. Evaluating the route lines at $m = a$ similarly proves this for P_i . Finite minima and maxima preserve strict increase. Finally, choosing a minimizing θ at the larger upper slope proves strict increase of C_A itself.

For the lower endpoint, use $\theta = 1, b = 2a$. Direct substitution gives a zero-terminal bound $(1-a)/(12(1-2a)) < a$, an early bound at most $(1+11a)/21$, and a late bound at most $(1+6a)/16$. The latter two are at most a , strictly so for $a > 1/10$. At $a = 1/10, b = 1/5, \gamma = 0$ forces the early bound to be exactly a for every θ , establishing equality there.

At $b = 1/2, \lambda = 0$, so all three intersection values equal L_0 . The other early bound is at least $E_0 = 4L_0/3$. The zero-terminal bound is at most $F = L_0$, while $P_\ell \geq L_0$ and $P_v \geq L_0$. Consequently $C_\theta(a, 1/2) = L_0$ for every θ . Now $L_0 \geq a$, with equality only at $a = 1/6$. The intermediate value theorem and strict increase give the unique root. Continuity of the root follows by taking convergent subsequences in the compact interval and using uniqueness, including the two endpoints. $\square$

This gives the new curve B_A stated at the beginning of the manuscript. It dominates B_M : on the latter's nontrivial middle range, $\theta = 0$ gives the same early bounds as V , no larger late bounds, and $Z \leq (1-q)/3 \leq \min(E_2, J_E)$. Thus $C_A \leq C_0 \leq V$. Below that range the previous curve is $2d-1$; above it the two displayed curves agree.

An exact new example, requiring no optimization, is

$$a = \frac{11}{100}, \quad b = \frac{9}{40}, \quad \theta = 1, \quad C_1(a, b) = \frac{644}{5925}, \quad a - C_1(a, b) = \frac{31}{23700} > 0.$$

The fully interpolated construction also covers $a = 7/50, b = 11/35, \theta = 1/3$, with cost $452437/3234750$ and margin $214/1617375$. These are rational checks of strict inequalities, not numerical evidence substituting for the profile or analytic proof.

6 Algebra and proof of the dimension criterion

Put $a = s - 1$ and $b = u - 1$. For $0 < a \leq 1/4$, the unforced cost $U(a, b)$ is continuous across $b = 1/2$. On each side it is increasing in b . The derivatives are respectively

$$\frac{a(1-a)}{2(2b-a)^2}, \quad \frac{1+2a}{(1+2b)^2}.$$

In the low-slope region, direct multiplication gives

$$U(a, b) < a \iff b(1 - 5a) < a(1 - 3a).$$

When $1/7 < a \leq 1/6$, this is precisely

$$b < \frac{a(1 - 3a)}{1 - 5a}.$$

The right side is at most $1/2$, because that comparison is equivalent to $(6a - 1)(a - 1) \geq 0$. In dimension variables its cutoff is $(2 + 2s - 3s^2)/(6 - 5s)$. Its improvement over $2s - 1$ follows from

$$\frac{a(1 - 3a)}{1 - 5a} - 2a = \frac{a(7a - 1)}{1 - 5a} > 0.$$

For $b \geq 1/2$, the high-slope condition is

$$\frac{b - a}{1 + 2b} < a \iff b < \frac{2a}{1 - 2a}.$$

For $1/6 < a \leq 1/4$ the right side exceeds $1/2$, and its dimension cutoff is $1/(3 - 2s)$. All $b \leq 1/2$ are covered in this range, since

$$U(a, b) \leq U(a, 1/2) = \frac{1 - 2a}{4} < a.$$

At $a = 1/6$, strict $b < 1/2$ also gives $U < a$. The joins at $d = 8/7$ and $d = 7/6$ have values $9/7$ and $3/2$, respectively.

Proof of Theorem 1.3. For $1 < d \leq 8/7$, apply the coherent criterion, Proposition 2.5, with both sets equal to E . For $8/7 < d \leq 5/4$, the preceding algebra shows that $D < B_U(d)$ implies $U(d - 1, D - 1) < d - 1$. The unforced finite-profile criterion, Theorem 3.1, gives the required pin. Its compact restrictions are subsets of E , so the pin lies in E . Strictness and continuity supply every exponent slack used in that theorem's Borel reduction. $\square$

Proof of Theorem 1.2. For $1 < d \leq 9/8$, use Proposition 2.5. For $9/8 < d < 7/6$, first use that proposition if $D < 2d - 1$. Otherwise put $a = d - 1$, $b = D - 1$. The root characterization in Section 4 gives $V(a, b) < a$, with $1/8 < a < 1/6$ and $2a \leq b < 1/2$. The profile estimate proved there supplies exactly the same bounded-length chains, admissibility, and perturbation errors as Lemma 3.3. Thus the proof of the finite-profile criterion applies with $C(s, u) = V(s - 1, u - 1)$. Explicitly, continuity and strictness allow $s < d$ and $u > D$, still with $1/8 < s - 1 < 1/6$, $2(s - 1) < u - 1 < 1/2$, and $V(s - 1, u - 1) < s - 1$. Use these parameters in the compact restriction, finite regularization, deletion, Fourier iteration, and shell summation of Section 3. The error and parameter choices are unchanged. For $7/6 \leq d \leq 5/4$, apply Theorem 1.3. $\square$

Proof of Theorem 1.1. For $1 < d \leq 11/10$, use Proposition 2.5. For $11/10 < d < 7/6$, the same proposition covers $D < 2d - 1$. Otherwise put $a = d - 1$, $b = D - 1$. Lemma 5.2 gives $C_A(a, b) < a$, and a minimizing parameter θ has $C_\theta(a, b) < a$. Keep this parameter fixed. Joint continuity and strictness allow $s < d$, $u > D$, with $1/10 < s - 1 < 1/6$, $2(s - 1) < u - 1 < 1/2$, and $C_\theta(s - 1, u - 1) < s - 1$. Use Lemma 5.1 in the finite-profile criterion, Theorem 3.1, choosing its fixed initial loss below $1/16$. The chain count, rounding error and approximate-barrier error are precisely those required there. The same compact restriction, finite regularization, deletion, Fourier iteration and shell summation give a pin in E . For $7/6 \leq d \leq 5/4$, use Theorem 1.3. $\square$

7 What the sharpness statement does and does not establish

The next proposition concerns only the scale profiles and the specified edge-cost model. It shows that a better limiting cost in the high-slope region needs information beyond the two linear barriers and Lipschitz condition. A different analytic method or additional geometric information could still weaken the dimension condition.

7.1 A sharp unforced profile obstruction above slope one half

The following example establishes sharpness only within the elementary profile problem. It is not a planar set with a null distance set and does not establish a necessary Hausdorff–packing condition.

Proposition 7.1 (Exact single-collapse cost). *For $0 < a < b \leq 1$, let*

$$g(x) = \min\{bx, 1 + a - x\} \quad (0 \leq x \leq 1), \quad C = \frac{b - a}{1 + 2b}.$$

This is a 1-Lipschitz profile with $ax \leq g(x) \leq bx$. For $0 < \zeta < C$, the infimum of the costs of all admissible chains from $1 - \zeta$ to zero, with no forced intermediate endpoint, is exactly $C - \zeta$. An attaining chain has $O(1 + \log(1/\zeta))$ edges.

Proof. The two affine pieces meet at

$$p = \frac{1 + a}{1 + b}.$$

Their slopes are b and -1 , so the Lipschitz assertion holds. The lower barrier on the terminal piece follows from $1 + a - x - ax = (1 + a)(1 - x) \geq 0$, and all other barrier comparisons are immediate. Put

$$s = 1 - C = \frac{1 + a + b}{1 + 2b}, \quad q = 2s - 1 = \frac{1 + 2a}{1 + 2b}.$$

The assumption $a < b$ gives $q < p < s$.

Consider any chain, and its first edge from $n > s$ to an endpoint $u \leq s$. Every preceding edge lies on the terminal affine piece of slope -1 , so the preceding costs sum to $1 - \zeta - n$. If $u \geq p$, the crossing edge costs $n - u \geq n - s$. If $u < p$, admissibility gives $u \geq 2n - 1 > q$. Since the profile has only one peak, its minimum over the crossing interval is at an endpoint, and the crossing cost is

$$(bu - (1 + a - n))_+ \geq (1 + 2b)(n - s).$$

In either case, the total cost through this crossing edge is at least $C - \zeta$. All subsequent costs are nonnegative, proving the lower bound.

For attainment, double the complementary depth from $1 - \zeta$ down to s , shortening the last jump if needed. This portion lies on the terminal descending piece and costs exactly $C - \zeta$. The next edge $s \rightarrow q$ is admissible with equality and has zero cost, because

$$g(s) = 1 + a - s = \frac{b(1 + 2a)}{1 + 2b} = g(q),$$

and the profile is above this common value between the endpoints. On $[0, q]$, the profile is increasing, so all downward edges cost zero. Maximal admissible jumps finish at zero without further cost. Their number, together with the initial doubling part, has the stated logarithmic bound. $\square$

For $b \geq 1/2$, the limit C in this example matches the universal upper bound in Lemma 3.3. Thus the limiting universal profile constant is sharp in that range. The example remains valid for $b < 1/2$, where equality with the universal minimax is not asserted. When $a = b$, the profile is simply ax and all costs are zero.

At $a = 1/4$, $b = 1$, the example has peak $p = 5/8$, critical endpoints $s = 3/4$, $q = 1/2$, and limiting cost $1/4$. It therefore reaches the strict source-margin boundary at the full-packing dimension pair $d = 5/4$, $D = 2$. This obstructs a uniform improvement of this particular profile constant. It does not rule out a different analytic argument, additional geometric information, or a different finite-scale balance, and it supplies no distance-set counterexample.

Proving a weakest possible condition for actual distance sets would require a necessity argument or counterexamples to positive-length pinned distances. None is supplied here. In particular, the sharp profile at $a = 1/4, b = 1$ explains why this method reaches the five-quarters boundary without crossing it; it does not establish the optimal Falconer threshold.

8 Plan for formal verification

No unconditional Lean verification or comparator acceptance is claimed. The existing target must retain the standard Euclidean distance, the same Borel-set hypotheses, and the standard Hausdorff and packing dimensions. No analytic conclusion may be inserted as an axiom or hidden in a definition.

The elementary part can be developed first: define finite admissible chains with only $2n - m \leq N$; prove zero-cost merging and its length bound; formalize the affine-potential construction, its localized low-slope version, and the perturbation and rounding bounds. Then prove the algebra of U and B_U , including strict joins and continuity. The minimum refinement additionally needs the tail-minimum reflection, the three affine intersections, their positive denominators and strict monotonicity, and existence and continuity of the unique root b_* . These are exact real-analysis statements; numerical root plots are not formal proofs of the curve. The adaptive extension additionally requires the edgewise endpoint-clipping inequality, the supporting-line stopping argument, and the variable free endpoint. Formalize the rational function C_θ before defining its minimum over the compact interval. Prove joint continuity, attainment, strict monotonicity in the upper slope and the root theorem. A minimizing parameter at the larger slope proves that compact minimization preserves strict increase here. The parameter is then fixed before choosing the strict Frostman and covering exponent slacks in the Borel reduction.

The analytic part needs independent formal proofs of Frostman extraction, finite regularization, averaged radial projection, packet localization, the inherited marking identities, local orthogonality, local constancy, conditional tube counting, the pinned quadratic identity, and the joint limit. The standard spatial packets and the finer spectral labels must be separate objects. Formalizing their exact sum identity before any inequality would catch the main potential gap in this refinement.

A useful order of deliverables is:

Milestone	Required checked statement
Profile calculus	A bounded admissible chain with the exact cost and explicit grid error.
Finite geometry	Cube ancestry, conditional normalization, tube overlap, and both angular inequalities.
Packet/spectral interface	Whole-standard-cap reconstruction and the exact inherited spectral sum.
Quadratic step	One tube threshold per inflation step, with every parameter dependency explicit.
Measure conclusion	Summable shell densities identify the original joint distance law and a positive-length pin.

Finally assemble the measure theorem and compact/Borel reduction, and compare the elaborated unconditional statement with a fixed reference statement. Build every dependency and inspect the theorem's axiom report. Acceptance requires only `propext`, `Classical.choice`, and `Quot.sound`; `sorryAx`, extra analytical axioms, or a conditional replacement target would fail the requested standard. This is a development plan, not evidence that the comparator has already accepted the theorem.

9 A weighted embedding for one Fourier step

This appendix proves the one-step estimate (29) directly from its geometric and Fourier hypotheses. In particular, the selected caps may depend arbitrarily on the child cube. The proof does not restore the omitted caps before applying the good-tube condition. The pin measure need not satisfy a Frostman bound in this step.

For a bounded rectangle A of positive area, write $m_A = |A|^{-1} \mathcal{L}^2|_A$. Dilations of cubes and rectangles below are concentric. We use the Fourier convention $\widehat{f}(\xi) = \int e^{-2\pi i x \cdot \xi} f(x) dx$.

Proposition 9.1 (Selected-cap weighted embedding). *Fix an integer $K \geq 1$, an index $0 \leq j < K$, and constants $C_0, B_0 \geq 1$. Let $0 < a \leq b$, and let P be an axis-parallel square of side b . Let $\mathcal{Q}$ be a finite subfamily of squares of an axis-parallel a -grid, each contained in P . For $L \geq L_0(C_0, K)$, set*

$$Q^+ = L^{2j+2}Q, \quad \widetilde{Q} = LQ^+, \quad P^+ = L^{2j+4}P, \quad D = L^{4K+20}.$$

Let σ be a finite positive Borel measure, and put

$$w_Q = \sigma(Q^+), \quad M = \sigma(P^+).$$

Let Θ be a finite label set. For every $\theta \in \Theta$ choose a function $f_\theta \in L^2(\mathbb{R}^2)$, a unit vector u_θ , a frequency center ξ_θ , and an effective unit direction v_θ , subject to the following conditions:

1. The Fourier support of f_θ is contained in

$$\left\{ \xi : |(\xi - \xi_\theta) \cdot u_\theta^\perp| \leq C_0/a, \quad |(\xi - \xi_\theta) \cdot u_\theta| \leq C_0/b \right\}.$$

These supports are also contained in balls of radius C_0/a whose pointwise overlap is at most B_0 .

2. The angular mismatch obeys

$$|u_\theta - v_\theta| \leq C_0 a/b.$$

Directions may first be given consistent signs, since the rectangles and the tests depend only on the unoriented lines.

3. For each child there are numbers $\varepsilon_{Q,\theta} \in \{0,1\}$. Let $T_{Q,\theta}$ be the rectangle centered at the center of Q , of width Da perpendicular to v_θ and length Db parallel to v_θ . For a number $H \geq 1$, assume only that

$$\varepsilon_{Q,\theta} = 1 \implies \sigma(T_{Q,\theta} \cap P^+) \leq H \frac{a}{b} M. \quad (\text{G})$$

The function f_θ is the same for all children; only the selection numbers depend on Q . Then, for every $m > 0$,

$$\begin{aligned} & \sum_{Q \in \mathcal{Q}} w_Q \int_{Q^+} \left| \sum_{\theta \in \Theta} \varepsilon_{Q,\theta} f_\theta \right|^2 dm_{Q^+} \\ & \leq CL^{8j+14} HM \sum_{\theta \in \Theta} \int_{P^+} |f_\theta|^2 dm_{P^+} + C_m M a^{-2} L^{-m} \sum_{\theta \in \Theta} \|f_\theta\|_2^2. \end{aligned} \quad (59)$$

Here C and C_m may depend on the fixed constants C_0, B_0, K , and C_m also on m . They do not depend on a, b, L, H , the measure, or the numbers of children and labels.

Proof. We separate the geometric consequence of the selected tests, local orthogonality, and the weighted reproducing-kernel estimate.

Geometry and the selected tube weights. The enlarged child cubes have overlap at most

$$\Lambda \leq CL^{4j+4}. \quad (60)$$

Indeed, in each coordinate at most $L^{2j+2} + 2$ grid centers can place a point in the corresponding enlarged interval. Also, for a point in $\tilde{Q}$, its distance from the center of P in the maximum norm is at most $(1 + L^{2j+3})b/2$. It follows that

$$\text{dist} \left(\bigcup_{Q \in \mathcal{Q}} \tilde{Q}, \mathbb{R}^2 \setminus P^+ \right) \geq Lb \quad (61)$$

for $L \geq 4$. In particular all Q^+ lie in P^+ , and

$$\sum_Q w_Q \leq \Lambda M. \quad (62)$$

Fix θ , and let U be any translate of a rectangle of width a and length b , oriented along u_θ . We claim

$$\sum_{\substack{Q: \varepsilon_Q, \theta=1 \\ \tilde{Q} \cap U \neq \emptyset}} w_Q \leq \Lambda H \frac{a}{b} M. \quad (63)$$

If the index set is empty this is immediate. Otherwise choose one of its cubes, Q_0 . Every Q^+ in that sum lies inside $T_{Q_0, \theta}$, as we now check.

Let q, q_0 be the centers of Q, Q_0 , and choose $z \in \tilde{Q} \cap U$, $z_0 \in \tilde{Q}_0 \cap U$. For $x \in Q^+$, write

$$x - q_0 = (x - q) + (q - z) + (z - z_0) + (z_0 - q_0).$$

The first, second, and fourth terms have length at most $CL^{2j+3}a$. The third has components at most a and b perpendicular and parallel to u_θ . Therefore

$$|(x - q_0) \cdot u_\theta^\perp| \leq CL^{2j+3}a, \quad |(x - q_0) \cdot u_\theta| \leq CL^{2j+3}b.$$

Changing to v_θ adds at most $CC_0L^{2j+3}a$ to the transverse component. The longitudinal component stays bounded by $C(1 + C_0)L^{2j+3}b$. Since $2j + 3 \leq 2K + 1$, the prescribed value of D exceeds twice these factors once $L \geq L_0(C_0, K)$. This proves the containment. All the cubes in question also lie in P^+ . Their overlap bound and the selected test at Q_0 now give

$$\sum_{\substack{Q: \varepsilon_Q, \theta=1 \\ \tilde{Q} \cap U \neq \emptyset}} \sigma(Q^+) \leq \Lambda \sigma(T_{Q_0, \theta} \cap P^+) \leq \Lambda H(a/b)M,$$

as claimed. Thus the mass bound for *every* dual rectangle in (63) is a consequence of the given tests at selected children, not an additional assumption.

Local orthogonality for a selected subset. Choose a Schwartz function v whose Fourier transform is supported in the unit ball and whose modulus is at least one on the unit square centered at zero. To obtain one, start with the inverse transform of a smooth bump supported in a sufficiently small ball; its value can be made nearly constant on that square and then multiplied by a fixed constant. For a child centered at q , put

$$v_Q(x) = v \left(\frac{x - q}{L^{2j+2}a} \right).$$

The product $v_Q f_\theta$ has Fourier support in the corresponding frequency ball enlarged by at most a^{-1} . These enlarged balls have overlap at most a fixed multiple of B_0 . In fact, dilating equal-radius balls by two increases their overlap by at most a factor nine in the plane: balls contributing at a point have their original balls inside the concentric ball of three times the radius, and an integration of the original overlap bound gives this count.

For every subset $A \subseteq \Theta$, Plancherel and pointwise Cauchy–Schwarz in frequency therefore give

$$\left\| v_Q \sum_{\theta \in A} f_\theta \right\|_2^2 \leq C \sum_{\theta \in A} \|v_Q f_\theta\|_2^2. \quad (64)$$

The constant is independent of the subset. Apply this to $A = \{\theta : \varepsilon_{Q,\theta} = 1\}$. Restrict the left side to Q^+ , and truncate the right side at $\tilde{Q} = LQ^+$. Schwartz decay and normalization of the cube averages give, for every $p > 0$,

$$\begin{aligned} \int_{Q^+} \left| \sum_{\theta} \varepsilon_{Q,\theta} f_\theta \right|^2 dm_{Q^+} \\ \leq CL^2 \sum_{\theta: \varepsilon_{Q,\theta}=1} \int_{\tilde{Q}} |f_\theta|^2 dm_{\tilde{Q}} + C_p a^{-2} L^{-p} \sum_{\theta} \|f_\theta\|_2^2. \end{aligned} \quad (65)$$

For the first term the area ratio is exactly L^2 . For the second, $|Q^+|^{-1} \leq a^{-2}$, and the square of the cutoff is bounded by $C_p L^{-p}$ outside $\tilde{Q}$. No unselected label has been introduced into the main term.

The anisotropic weighted embedding. For a fixed label define the finite positive averaging measure

$$\tau_\theta = \sum_{Q: \varepsilon_{Q,\theta}=1} w_Q m_{\tilde{Q}}.$$

Equations (63) and (62) imply

$$\tau_\theta(U) \leq \Lambda H(a/b)M \quad \text{for every dual rectangle } U, \quad \tau_\theta(\mathbb{R}^2) \leq \Lambda M. \quad (66)$$

Take a smooth compact frequency bump equal to one on $[-C_0, C_0]^2$. Translating, rotating, and rescaling it produces a reproducing kernel k_θ for the Fourier rectangle in the proposition. Thus $f_\theta = f_\theta * k_\theta$, and

$$\begin{aligned} \|k_\theta\|_1 &\leq C, \\ |k_\theta(z)| &\leq \frac{C_p}{ab} \left(1 + \frac{|z \cdot u_\theta^\perp|}{a} + \frac{|z \cdot u_\theta|}{b} \right)^{-p} \quad (p > 0). \end{aligned} \quad (67)$$

The translation of the frequency rectangle merely modulates the kernel and does not change these absolute-value estimates. Each L^2 function with the specified bounded Fourier support has a continuous representative, since its Fourier transform is also integrable. The reproducing identity and the following pointwise estimate are understood for that representative. Cauchy–Schwarz yields

$$|f_\theta(x)|^2 \leq C \int |f_\theta(y)|^2 |k_\theta(x-y)| dy.$$

For each fixed y , partition the plane into translates of dual rectangles a -by- b centered on the corresponding rectangular lattice based at y . The mass estimate in (66), the kernel bound with any $p > 2$, and summation of the lattice decay give

$$\int |k_\theta(x-y)| d\tau_\theta(x) \leq \frac{C}{ab} \Lambda H \frac{a}{b} M = C \frac{\Lambda H M}{b^2}. \quad (68)$$

This computation displays the single occurrence of the tube threshold.

If $y \notin P^+$, the support margin (61) gives $|x - y| \geq Lb$ on the support of τ_θ . Since $a \leq b$, the anisotropic distance in (67) is then at least a fixed multiple of L . Its arbitrary-order decay and the total mass in (66) give the additional bound

$$\int |k_\theta(x - y)| d\tau_\theta(x) \leq C_p \frac{\Lambda M}{ab} L^{-p} \quad (y \notin P^+).$$

Integrating the pointwise estimate for f_θ , and using Tonelli's theorem, now proves

$$\int |f_\theta|^2 d\tau_\theta \leq C \frac{\Lambda H M}{b^2} \int_{P^+} |f_\theta|^2 dy + C_p \frac{\Lambda M}{ab} L^{-p} \|f_\theta\|_2^2. \quad (69)$$

Finally sum (65) with weights w_Q , and apply (69) to each label. The main coefficient, after converting to the normalized parent average, is

$$CL^2 \Lambda H M \frac{|P^+|}{b^2} \leq CL^{2+4j+4+4j+8} H M = CL^{8j+14} H M.$$

The local-orthogonality remainder is bounded by $C_p \Lambda M a^{-2} L^{-p} \sum_\theta \|f_\theta\|_2^2$. The weighted-kernel remainder has the same bound with an additional L^2 , since $(ab)^{-1} \leq a^{-2}$. Choosing the arbitrary decay order p larger than $m + 4j + 6$, and absorbing (60), gives the stated remainder. This proves the proposition. $\square$

Verification of the hypotheses in the unforced transfer

Fix a spatial parent $P = Q_{j+1}$ and one parent angular label in Section 3.5. Apply Proposition 9.1 with

$$a = a_j = r_j/R, \quad b = a_{j+1} = r_{j+1}/R, \quad L = R^h, \quad \sigma = \sigma_i, \quad H = H_{i,j}.$$

The spatial dilations and the test dilation then coincide exactly with those specified there, with $K = K_0$. In particular the proposition does not require replacing the current tests by tests on additional parents or by a new measure.

Take

$$f_\theta = F_{P,\theta}, \quad \varepsilon_{Q,\theta} = \mathbf{1}_{\{\theta \text{ is good for } Q\}}.$$

The inherited identity (28) says that each child's function is exactly the selected sum in the proposition. For this fixed parent the functions $F_{P,\theta}$ do not depend on the child. The selections are constants on the averaging cubes, even though different cubes may choose completely different cap subsets.

The actual Fourier direction u_θ is the direction of the auxiliary angular interval, and v_θ is its effective marking direction. Their difference is $O(\bar{\delta}_j)$, with $\bar{\delta}_j \leq a_j/a_{j+1}$ by (24). This supplies the angular mismatch hypothesis. The marking rule supplies (G): multiplying its conditional probability bound by $\sigma_i(P^+)$ gives precisely the unnormalized inequality there. Thus the stronger-looking bound (63) for arbitrary dual rectangles has been derived from the existing marks, rather than silently assumed.

The circular support of each auxiliary cap has tangential frequency width $O_T(R/r_j) = O_T(a^{-1})$ and normal width $O_T(R/r_j^2) \leq O_T(R/r_{j+1}) = O_T(b^{-1})$. The fixed frequency convolution adds bounded width, absorbed because $b \leq 1$. The supports lie in bounded-overlap balls of radius $O_T(a^{-1})$. When labels are pairs (S, I) , only boundedly many standard labels S are nonzero at a direction, and the dyadic intervals have bounded overlap. Their enclosing balls have uniformly bounded overlap as well: consecutive angular centers at a given level are separated by a fixed

multiple of δ_j , up to this bounded multiplicity. The constants C_0, B_0 may depend on the fixed annular ratio, hence on T , but not on R . The terminal widths are at least a constant times R^{-1} , so the fixed frequency smoothing causes no growing ball multiplicity.

All functions in this application belong to L^2 . For example, a bounded-density finite circular measure convolved with the fixed compact smooth frequency function has a smooth compactly supported Fourier representative. This observation applies also to the hard angular indicator pieces; no differentiation of those indicators is needed anywhere in Proposition 9.1.

Consequently (59), followed by summation over parents and parent angular labels, proves (29), with a loss $R^{C_{\kappa_0} h}$ and the single factor $H_{i,j}$. The remainder is rapidly decreasing in the required sense. Indeed $a_j \geq R^{-1}$, and for any target decay power A , the remainder in the proposition contains

$$a_j^{-2} L^{-m} \leq R^{2-hm}.$$

For fixed $h > 0$, choose m sufficiently large to absorb A and the polynomial bounds on the remaining sums and global norms. Constants depending on this choice are independent of the frequency.

Remark 9.2 (Scope and necessary distinctions). The proposition permits arbitrary finite positive pin measures, even atomic ones, and arbitrary child-dependent subsets of the fixed cap family. It would not justify allowing the selection to vary pointwise inside a child average, substituting unrelated numbers for the masses $\sigma(Q^+)$, or replacing the common parent functions by child-dependent functions. The present transfer makes none of those changes. This appendix verifies the one-step analytic implication; it does not establish optimality of the profile bound or of the resulting dimension condition.

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